Forgetting as Interference in Continual Learning: Optimal Merging and Interference-Gated Allocation

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Abstract

Continual learning commonly relies on post-hoc mechanisms such as replay, elastic regularization, or distillation, which mitigate interference only after it has occurred. This work instead models interference directly. In frozen-feature, linear heads, forgetting equals interference energy $\frac{1}{2}\Delta^\top \Sigma_A \Delta$ induced by a later update Δ, measured in the earlier task's input covariance Σ_A. This identity extends to deep networks by measuring curvature along the path of the task loss, with only a small number of additional forward passes.

Two consequences follow. First, interference is removable when task supports are disjoint ($\Delta \in \ker \Sigma_A$); otherwise a non-zero distortion floor is unavoidable and is given by the posterior-variance-weighted target gap, with corresponding mutual-information bounds. Second, optimal model merging reduces to Σ-orthogonalization of task vectors, and continual learning can be interpreted as its online greedy analogue. Based on this view, this work introduces Interference-Gated Functional Allocation (igfa), a replay-free, Fisher-free method that shares capacity when a regime-specific task similarity exceeds $s^* \approx 0.26$ and orthogonalizes updates under conflict.

Empirically, igfa yields lossless retention on disjoint supports and shifts unavoidable forgetting into deferred but recoverable plasticity. On Split-Digits it matches or exceeds strong replay-free and Fisher-free baselines, achieving 0.98 accuracy with near-zero forgetting using only a low-rank state summary. On Split-CIFAR-100 with a frozen ViT it reaches parity with the strongest structural baseline at the best retention level, without replay or Fisher information. At billion-parameter scale with `pythia-1b`, continual pretraining shows the same replay-free, Fisher-free retention mechanism achieving the lowest loss and the least forgetting, while a pre-flight diagnostic correctly predicts when retention is feasible.

Keywords: continual learning; catastrophic forgetting; model merging; neural tangent kernel; gradient projection; rate–distortion; representation geometry

1 Introduction

A model trained sequentially on task B after task A often degrades on task A, a phenomenon known as catastrophic forgetting. Networks trained by stochastic gradient descent (SGD) or its variants update from the current minibatch alone (or a smoothed average of a short window of minibatches); the update is therefore *oblivious to past knowledge.* This obliviousness is desirable when the training data are i.i.d., but harmful once the training distribution shifts over time in the continual setting. Standard approaches such as replay [3, 4, 5], elastic weight consolidation [1, 2], and distillation [6] mitigate this effect after interference has occurred. Although effective, these methods are primarily corrective: replay requires data storage and may raise privacy concerns, regularization constrains plasticity, and none directly characterizes when interference is unavoidable and when it can be eliminated.

The analysis focuses on the regime in which a frozen feature extractor $\phi(x)$ is paired with a trainable linear head, covering the common frozen-backbone and parameter-efficient fine-tuning (PEFT) settings and the first-order neural tangent [11] approximation of a fully trained network. The forgetting of an earlier task A after an update Δ caused by learning task B is exactly the interference energy $\frac{1}{2}\Delta^\top \Sigma_A \Delta$, where $\Sigma_A = \mathbb{E}_{x\sim D_A}\left[\phi(x)\phi(x)^\top\right]$ denotes the feature second moment of task A, measuring which feature directions are active for that task. The same geometry predicts per-task forgetting accurately and extends approximately to deeper networks with feature drift.

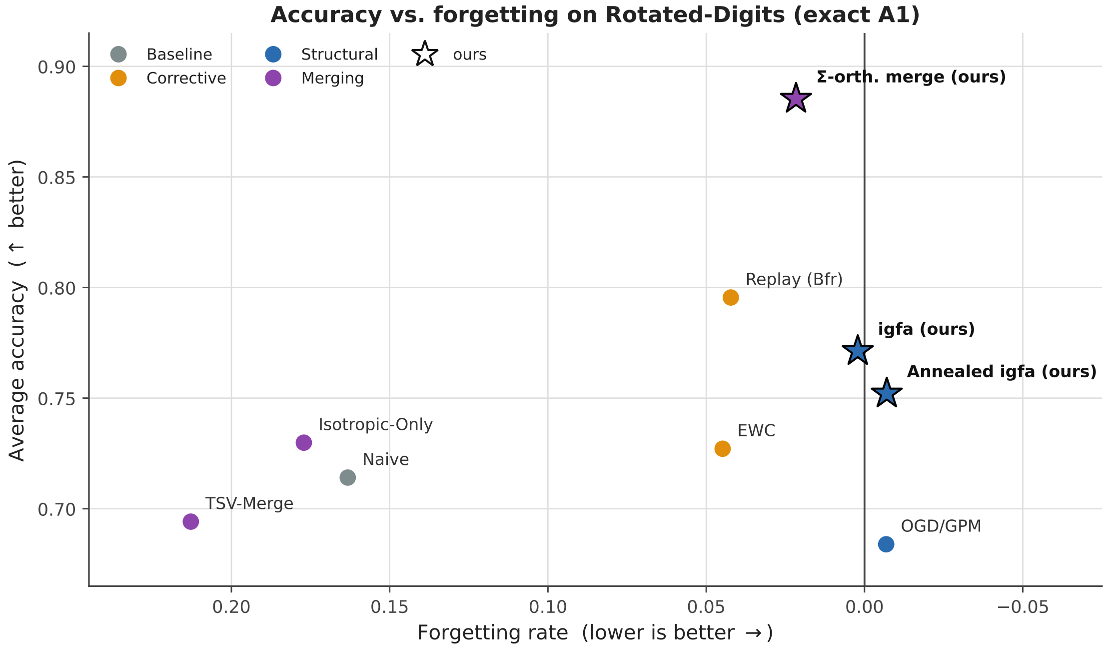

Figure 1: Accuracy versus forgetting. Nine methods on the exact-A1 Rotated-Digits benchmark (five rotations, frozen random features; mean over 5 seeds) are differentiated by method class (baseline / corrective / structural / merging); our methods are marked "(ours)"; higher accuracy (up) and lower forgetting (right) are better. *Corrective* (EWC, replay) and *online structural* (OGD/GPM, igfa) methods learn the stream sequentially, whereas *merging* methods solve each rotation independently and combine offline. *(i)* The Σ-orthogonal merge of Theorem 3 attains near-oracle accuracy (0.885 against a 0.903 per-task ceiling) with the smallest post-merge residual floor ($D = 0.032$, versus 0.20 and 0.38 for the isotropic and Euclidean merges), confirming that the task-induced, non-Euclidean Σ-metric is the correct inner product for merging. *(ii)* Among online, buffer- and Fisher-free methods, our signed gate (igfa) attains the highest accuracy; annealing the protection boundary (Annealed igfa) trades slightly lower accuracy for lower, even negative, forgetting. Full numbers with confidence intervals and the residual-D ablation are in Sec. S4.

The functional has three consequences. First, the geometry of Σ_A only penalizes components in its active subspace and does not change task A's loss for updates in the null space $\ker \Sigma_A$. Interference is removable for disjoint task supports (reduces to ordinary orthogonality in isotropic geometry), whereas overlapping supports imply a non-zero distortion floor. Second, the same functional gives an optimality condition for model merging: task vectors should be orthogonal under the Σ-induced metric, which makes offline merging and online continual learning two scheduling variants of the same objective. Third, it motivates Interference-Gated Functional Allocation (igfa), an online method that uses a low-rank summary of the occupied function subspace to decide whether a new update should be shared or orthogonalized (Fig. 2).

The paper makes three contributions.

1. **Interference functional and removability.** Section 3 shows that forgetting is exactly the interference energy induced by a later update. This yields a structural criterion for lossless retention and identifies the unavoidable distortion floor when task supports overlap. The same quantity admits an information-theoretic interpretation through the task-inference problem.

2. **Optimal merging as Σ-orthogonalization.** Section 3.4 shows that post-merge loss decomposes into task-wise fit and cross-task interference. Minimizing this objective requires mutual Σ-orthogonality of the task vectors, linking model merging and continual allocation under a common geometric principle.

3. **Interference-Gated Functional Allocation.** Section 4 introduces igfa, a replay-free and Fisher-free online algorithm that stores only a low-rank subspace summary. Unlike unconditional orthogonal projection methods, igfa shares capacity when tasks are sufficiently aligned and orthogonalizes updates only when interference is predicted.

All formal results are established in the linear-on-features regime. The experiments in Section 5 are designed to test the theory directly, including lossless allocation on disjoint supports, the relocation of unavoidable cost from forgetting into deferred plasticity, the similarity threshold at $s^* \approx 0.26$, and the onset of a capacity floor once the cumulative occupied rank reaches the feature dimension, $T\,r \approx d$ (with T

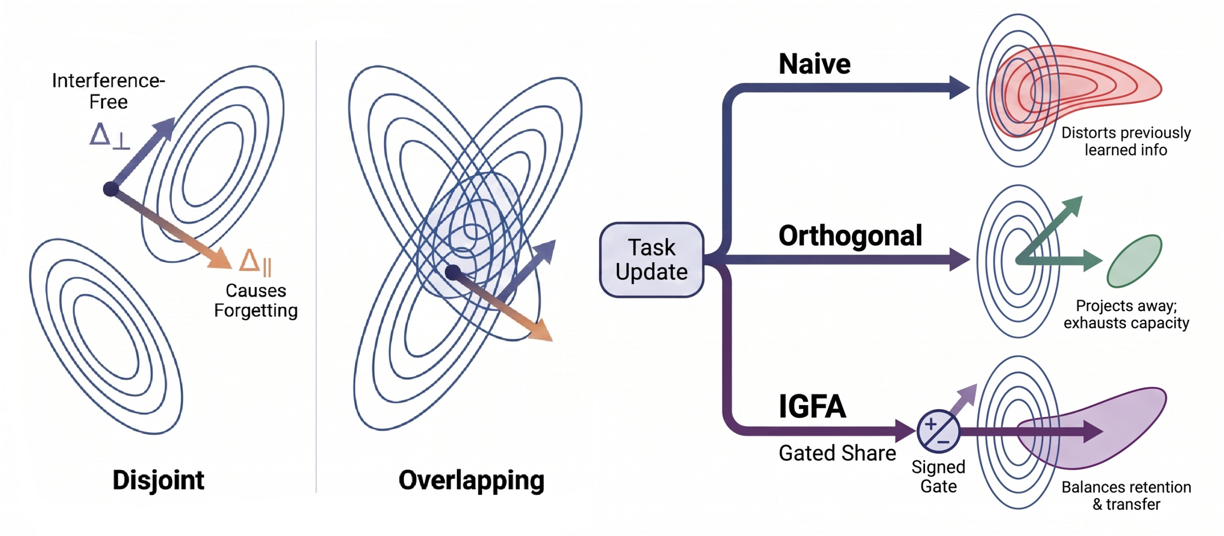

Figure 2: The task-induced geometry in feature space motivates either full sharing, full separation, or selective parameter sharing. *Left (Disjoint).* When two tasks excite orthogonal feature subspaces, the update $\Delta = \Delta_\parallel + \Delta_\perp$ splits into a component $\Delta_\parallel \in \operatorname{range} \Sigma_A$ across the iso-loss contours (the only part that costs forgetting) and $\Delta_\perp \in \ker \Sigma_A$ (interference-free); allocation is lossless. *Centre (Overlapping).* When the supports share an active direction, no update for B can avoid $\operatorname{range} \Sigma_A$ and a distortion floor is unavoidable. *Right.* The same functional yields three update rules. Naive descent incurs the full interference (distorts old knowledge); orthogonal projection (OGD/GPM [7, 8]) removes all overlap (no forgetting, but no transfer and capacity exhaustion); igfa applies a *signed* gate—sharing aligned directions and orthogonalizing conflicting ones—balancing retention and transfer while carrying only the occupied subspace as state.

the number of tasks in the stream, r the rank of the feature subspace each task occupies, and d the feature dimension). In the frozen-feature Split-Digits setting, igfa matches the strongest replay-free and Fisher-free baseline at about 0.98 accuracy with near-zero forgetting while using no replay buffer. Extensions to full-network training, large-scale overlays, and direct comparison with replay at scale are discussed in Section 6, with the supporting experiments detailed in the Supporting Information.

2 Related work

Corrective continual learning. Replay, regularization, and distillation remain the main corrective approaches to continual learning [3, 4, 5, 1, 2, 6]. They are effective baselines, but they do not explicitly model interference through task geometry. In contrast, the interference functional derived below predicts the collision and identifies when it can be removed before it occurs; replay can be viewed as a stochastic Monte-Carlo estimator of the same task second-moment Σ_t that we summarize directly via a low-rank subspace.

Geometry- and gradient-based methods. A related line of work controls the update direction itself. Orthogonal Gradient Descent [7], Gradient Projection Memory [8], and Orthogonal Weight Modification [9] project updates away from subspaces associated with previous tasks; in the present setting this corresponds to enforcing $\Delta \in \ker \Sigma_A$. PCGrad-style methods [10] resolve conflicts through pairwise gradient comparisons rather than explicit task-subspace protection. Recent work also studies closely related regimes: Geodesic-Aligned Gradient Projection [27] addresses feature drift during sequential learning, and MINGLE [28] uses a null-space criterion to gate low-rank experts during test-time merging. Relative to these approaches, the present framework derives the projection condition from an exact forgetting functional, replaces unconditional projection with a similarity-based share-or-orthogonalize rule, and uses the same functional to characterize both merging and the distortion floor. NTK-based analyses [12, 13] likewise relate forgetting to overlap in representation space, but do not develop the Σ-orthogonality criterion or the associated transfer gate.

Model merging and task arithmetic. A parallel literature merges independently fine-tuned models through arithmetic on weight-space task vectors [14, 15]. These methods usually operate in Euclidean parameter space, whereas the present analysis identifies the task-dependent metric induced by Σ_t as

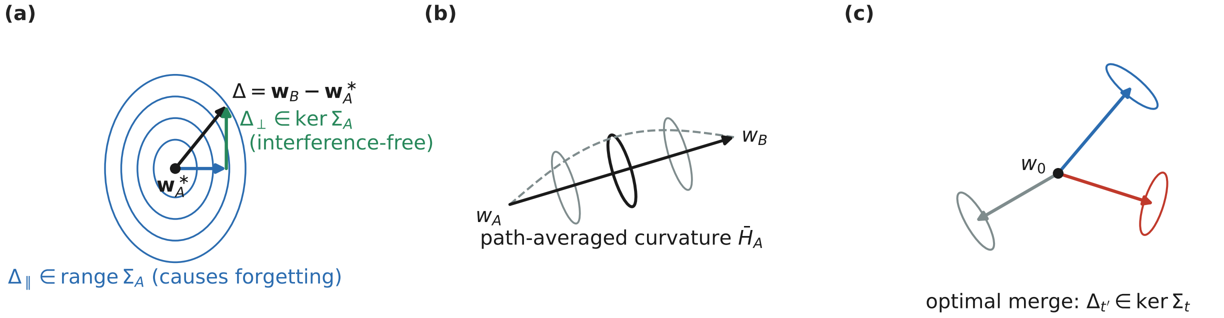

Figure 3: Geometry of the core identities. *(a)* Forgetting is the interference energy $\frac{1}{2}\Delta^\top \Sigma_A \Delta$; only the component $\Delta_\parallel \in \mathrm{range}\, \Sigma_A$ costs forgetting, while $\Delta_\perp \in \ker \Sigma_A$ is free (Theorem 1). *(b)* For general architectures the input covariance is replaced by the path-averaged curvature $\bar{H}_A$ along the segment $w_A \to w_B$ (Theorem 2). *(c)* Optimal merging makes the task vectors mutually Σ-orthogonal (Theorem 3).

the relevant geometry. Under this metric, optimal merging becomes mutual Σ-orthogonalization, with ordinary weight-space orthogonality recovered only when $\Sigma_t \propto I$. Task Singular Vectors [24] promotes orthogonality by whitening singular directions of task vectors; by contrast, isotropic merging [25] shows that shared-subspace isotropy can outperform strict orthogonalization. WUDI-Merging [23] is particularly close in spirit because it links task vectors to input subspaces and minimizes interference without data; the key difference here is that the metric and residual are *derived* from an exact forgetting functional rather than an approximate subspace argument. Tangent-space continual merging [26] also operates in a first-order regime, but with similar trade-offs between scalability and functional specificity. The common perspective adopted here is that the same interference functional governs both offline merging and online continual allocation.

Scenarios, scale, and plasticity. The standard task-, domain-, and class-incremental taxonomy [16] distinguishes whether task identity is given or must be inferred. In this context, the distortion floor developed later provides a quantitative link between support overlap and task inference. Related empirical and analytical work has examined the effects of scale and pretraining [17], frozen-backbone prompt and adapter methods [18, 19, 20], forgetting in linear models [21], plasticity loss [22], and convergence or task ordering in continual linear classification [29, 30].

3 Preliminaries: the interference functional

This section fixes the setting and assumptions (Sec. 3.1), derives the interference functional and the removability dichotomy (Sec. 3.2), generalizes the identity beyond frozen features (Sec. 3.3), and derives the optimal merge together with the distortion floor (Sec. 3.4).

3.1 Setting and assumptions

Let $\phi : \mathcal{X} \to \mathbb{R}^d$ be a fixed feature map and $w \in \mathbb{R}^d$ a trainable linear head with prediction $\hat{y} = w^\top \phi(x)$. For task t, let $\Sigma_t = \mathbb{E}_{x\sim\mathcal{D}_t}[\phi(x)\phi(x)^\top] \succeq 0$ denote the feature second moment, and let w_t^* be a realizable optimum. The per-task excess loss is

$$L_t(w) = \tfrac{1}{2}\,(w - w_t^*)^\top \Sigma_t\,(w - w_t^*)\ \geq 0. \tag{1}$$

Equation (1) is exactly the excess population risk under squared loss. Writing $y = \phi(x)^\top w_t^* + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\varepsilon \perp \phi(x)$ gives $\mathbb{E}[(\hat{y} - y)^2]/2 = L_t(w) + \frac{1}{2}\mathbb{E}[\varepsilon^2]$, so all loss differences below depend only on $L_t(w)$. Three assumptions fix the regime: **(A1) frozen features**—ϕ is fixed during training, exact for frozen-backbone and PEFT settings and the first-order NTK [11] model of end-to-end training, so full-network claims are first-order and are tested in Sec. 5.4; **(A2) realizability**—each w_t^* attains zero excess loss, so mis-specification adds only a task-constant and leaves the interference terms untouched; and **(A3) the quadratic regime**—training minimizes Eq. (1) by (projected) gradient descent, so motion is confined to the excited subspace while directions in $\ker \Sigma_t$ leave L_t unchanged.

3.2 The functional and the removability dichotomy

Train task A from a base point w_0 to convergence. Under A3, gradient descent moves only in $\mathrm{range}\, \Sigma_A$, so the converged solution w_A satisfies $L_A(w_A) = 0$ and equals w_A^* up to an arbitrary component in $\ker \Sigma_A$.

For loss differences we therefore take $w_A = w_A^*$. Now train task B, reaching w_B, and define $\Delta = w_B - w_A^*$.

Theorem 1 (Forgetting is interference energy). *Under A1–A3, the forgetting of task A caused by learning B is*

$$\Delta L_A = L_A(w_B) - L_A(w_A^*) = \tfrac{1}{2}\,\Delta^\top \Sigma_A\,\Delta, \qquad \Delta = w_B - w_A^*. \tag{2}$$

Proof. Since $L_A(w_A^*) = 0$ by A2, substituting $w_B = w_A^* + \Delta$ into Eq. (1) yields $L_A(w_B) = \frac{1}{2}\Delta^\top \Sigma_A \Delta$. No approximation is involved; the only essential assumption is that ϕ remains fixed, so Σ_A is unchanged before and after learning B. ☐

Decompose the update as $\Delta = \Delta_\parallel + \Delta_\perp$, with $\Delta_\perp \in \ker \Sigma_A$ and $\Delta_\parallel \in \operatorname{range} \Sigma_A$. Then $\Delta L_A = \frac{1}{2}\Delta_\parallel^\top \Sigma_A \Delta_\parallel$, so only the component of the update lying in task A's active subspace contributes to forgetting (Fig. 3a).

Corollary 1 (Removability dichotomy). *An update is lossless for A if and only if* $\Delta \in \ker \Sigma_A$. *Consequently there exists a way to learn B with zero forgetting of A if* w_B^* *is reachable through* $\ker \Sigma_A$, *i.e. if* $w_B^* - w_A^* \in \ker \Sigma_A$. *Even when this fails,* L_B *can still be strictly reduced without forgetting whenever* $w_B^* - w_A^*$ *has a non-zero component in* $\ker \Sigma_A \cap \operatorname{range} \Sigma_B$. *If* $\operatorname{range} \Sigma_A$ *and* $\operatorname{range} \Sigma_B$ *share an active direction on which* w_A^* *and* w_B^* *disagree, then no lossless update reaches* w_B^**, and a positive floor remains (Theorem 4).*

Proof. Since $\Sigma_A \succeq 0$, $\Delta^\top \Sigma_A \Delta = 0 \Leftrightarrow \Sigma_A^{1/2}\Delta = 0 \Leftrightarrow \Delta \in \ker \Sigma_A$. Projected gradient descent on L_B restricted to $w_A^* + \ker \Sigma_A$ converges to the minimizer on that affine set. It reaches w_B^* if $w_B^* - w_A^* \in \ker \Sigma_A$, and it strictly reduces L_B whenever the projection of $-\nabla L_B(w_A^*)$ onto $\ker \Sigma_A$ is non-zero, equivalently whenever $\Sigma_B(w_B^* - w_A^*)$ has a non-zero component in $\ker \Sigma_A$. In the disjoint-support case, $\operatorname{range} \Sigma_B \subseteq \ker \Sigma_A$, so the entire update is lossless. ☐

Here, "task support" refers to the column space $\operatorname{range} \Sigma_t$, that is, the subspace of feature directions activated by task t. Disjoint supports therefore mean orthogonal feature subspaces, not disjoint label sets.

3.3 The architecture-general identity

Theorem 1 assumes frozen features. In end-to-end training, feature drift degrades the frozen-curvature prediction, especially in deeper networks and at larger step sizes. The correct generalization is to replace Σ_A by the curvature of L_A averaged along the displacement from w_A to w_B.

Theorem 2 (Architecture-general forgetting identity). *Let* L_A *be any twice-differentiable task-A loss, and let training on B move the parameters from* w_A *to* w_B. *Writing* $\Delta = w_B - w_A$,

$$\Delta L_A = \nabla L_A(w_A)^\top \Delta + \tfrac{1}{2}\,\Delta^\top \bar{H}_A\,\Delta, \qquad \bar{H}_A = 2\int_0^1 (1-t)\,\nabla^2 L_A(w_A + t\Delta)\,dt, \tag{3}$$

where $\bar{H}_A$ *is the Hessian of* L_A *averaged along the segment* $w_A \to w_B$. *If* w_A *is a stationary point, then* $\nabla L_A(w_A) = 0$ *and* $\Delta L_A = \frac{1}{2}\Delta^\top \bar{H}_A \Delta$.

Intuition: a second-order Taylor expansion of L_A along the segment $w_A \to w_B$, with exact integral remainder, produces the gradient term and the path-averaged Hessian (full derivation in Sec. S1).
Theorem 1 is the constant-curvature special case: if $\nabla^2 L_A \equiv \Sigma_A$ then $\bar{H}_A = \Sigma_A$. For squared loss, $\nabla^2 L_A(w) = \mathbb{E}_{\mathcal{D}_A}[J_w J_w^\top] + R(w)$, so $\bar{H}_A$ is a path-averaged Gauss–Newton geometry, where $J_w = \partial f(x;w)/\partial w$ is the parameter Jacobian and $R(w)$ is a residual-curvature term (Fig. 3b). The removability, merging, and floor results therefore extend by replacing Σ_A with $\bar{H}_A$. In practice, $\bar{H}_A$ can be estimated from a small number of evaluations of L_A along the segment, without forming the Hessian explicitly. The same prescription tells igfa what to track under drift: the running Gauss–Newton subspace $\mathbb{E}[J_w J_w^\top]$ rather than a stale Σ_A.

3.4 Optimal Merging and the Distortion Floor

Represent K task solutions as $\Delta_t = w_t^* - w_0$ from a shared base w_0, and merge them as $\theta^* = w_0 + \sum_t \Delta_t$. Substituting into Eq. (1), the loss incurred on task t is

$$L_t(\theta^*) = \tfrac{1}{2}\Big(\textstyle\sum_{t' \neq t} \Delta_{t'}\Big)^\top \Sigma_t \Big(\textstyle\sum_{t' \neq t} \Delta_{t'}\Big), \tag{4}$$

that is, the interference energy of the other task vectors measured in task t's geometry.

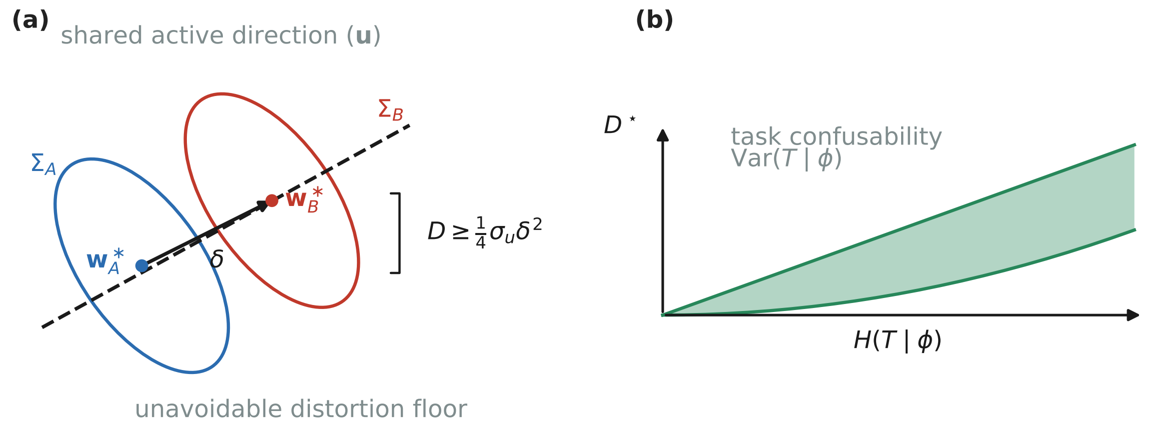

Figure 4: The distortion floor and its information-theoretic reading. *(a)* A floor appears only on a shared active direction $u \in \operatorname{range}\Sigma_A \cap \operatorname{range}\Sigma_B$ on which the targets disagree by δ, with $\mathcal{D} \geq \frac{1}{4}\sigma_u\delta^2$ (Theorem 4). *(b)* The Bayes single-head excess risk $\mathcal{D}^\star$ factors into the geometric target gap $(\phi^\top\delta)^2$—the displacement δ in (a)—and the task-confusability factor $\operatorname{Var}(T \mid \phi)$, which is two-sidedly controlled by the conditional entropy $H(T \mid \phi)$ (Theorem 5; band).

Theorem 3 (Optimal merging is Σ-orthogonalization)**.** *The total merge loss $\sum_t L_t(\theta^*)$ vanishes whenever the task vectors are pairwise Σ-orthogonal, i.e. $\Delta_{t'} \in \ker\Sigma_t$ for all $t' \neq t$; this condition is also generically necessary. When it fails, the minimum total loss over shared heads is attained at $\theta_J = \big(\sum_t \Sigma_t\big)^+\big(\sum_t \Sigma_t w_t^*\big)$, and for two tasks*

$$w_J = (\Sigma_A + \Sigma_B)^+(\Sigma_A w_A^* + \Sigma_B w_B^*). \tag{5}$$

The residual $\mathcal{D} := \sum_t L_t(\theta_J)$ is the distortion floor*.*

Proof. Since each $L_t(\theta^*) \geq 0$, the sum vanishes iff every term vanishes, i.e. $\Sigma_t^{1/2}\sum_{t'\neq t}\Delta_{t'} = 0$. A sufficient, and generically necessary, condition is $\Delta_{t'} \in \ker\Sigma_t$ for all $t' \neq t$. The minimizer of $\sum_t L_t(\theta) = \frac{1}{2}\sum_t(\theta - w_t^*)^\top\Sigma_t(\theta - w_t^*)$ satisfies $\sum_t \Sigma_t(\theta - w_t^*) = 0$, which gives θ_J; Eq. (5) is the two-task case. ☐

The relevant inner product for merging is therefore the task-dependent Σ_t-metric rather than the Euclidean metric of parameter space (Fig. 3c); Euclidean orthogonality is the white-feature special case $\Sigma_t \propto I$. The same residual $\mathcal{D}$ also governs the online continual-learning setting, so merging and continual learning optimize the same functional on different schedules.

When orthogonality fails: the distortion floor. Optimal merging requires Σ-orthogonality; when the supports force it to fail, the residual $\mathcal{D}$ of Theorem 3 does not vanish, and it has a task-inference reading. Consider two tasks as a single problem in which each input carries a hidden label $T \in \{A, B\}$, and a shared head must predict the task-appropriate target.

Theorem 4 (Distortion floor and inference obstruction)**.** *Suppose the tasks share an active direction u, with $\|u\| = 1$ and $u \in \operatorname{range}\Sigma_A \cap \operatorname{range}\Sigma_B$, and let $\delta = u^\top(w_B^* - w_A^*)$. Then the distortion floor obeys the lower bound*

$$\mathcal{D} \;\geq\; \tfrac{1}{4}\,\sigma_u\,\delta^2, \qquad \sigma_u = \big(u^\top(\Sigma_A^{-1} + \Sigma_B^{-1})u\big)^{-1}, \tag{6}$$

where Σ_A^{-1} and Σ_B^{-1} denote the inverses of Σ_A and Σ_B restricted to the shared active subspace $S =$ $\operatorname{range}\Sigma_A \cap \operatorname{range}\Sigma_B$. On the shared support, the excess Bayes risk of the Bayes-optimal single head over the task oracle equals the merge residual.

Intuition: restricting the merge to the shared direction u reduces it to a scalar two-point problem whose Σ-weighted optimum leaves residual $\frac{1}{4}\sigma_u\delta^2$; summing over shared directions and identifying the Bayes single head with the posterior mean gives the bound and the inference claim (Sec. S1).
Support overlap is therefore the common obstruction to both Σ-orthogonal merging and task inference; a positive floor appears only on shared active directions where the tasks also disagree. This yields the geometric counterpart of the task- versus class-incremental distinction [16]: disjoint supports make the task identity effectively free, whereas overlapping supports can induce a non-zero floor when the tasks disagree on shared active directions.

Theorem 5 (Information–estimation identity for the floor)**.** *Let $T \sim \mathrm{Unif}\{A, B\}$, $\phi \mid T \sim \mathcal{D}_T$, $y = \phi^\top w_T^* + \varepsilon$, with $\varepsilon \perp (T, \phi)$, $\mathbb{E}\varepsilon = 0$, and $\mathbb{E}\varepsilon^2 = \sigma^2$. Writing $\eta(\phi) = \Pr(T{=}A \mid \phi)$ and $\delta = w_A^* - w_B^*$, the excess Bayes risk of a single shared head over the task oracle is*

$$\mathcal{D}^\star = \tfrac{1}{2}\,\mathbb{E}_\phi\big[\,\mathrm{Var}(T\,|\,\phi)\,(\phi^\top\delta)^2\big] = \tfrac{1}{2}\,\mathbb{E}_\phi\big[\eta(1-\eta)\,(\phi^\top\delta)^2\big]. \tag{7}$$

With $h_b(p) = -p\ln p - (1-p)\ln(1-p)$ the binary entropy in nats, this implies the two-sided bounds

$$\tfrac{1}{8}\,\mathbb{E}_\phi\big[(\phi^\top\delta)^2\,h_b(\eta)^2\big] \;\le\; \mathcal{D}^\star \;\le\; \tfrac{1}{4}\,\mathbb{E}_\phi\big[(\phi^\top\delta)^2\,h_b(\eta)\big]. \tag{8}$$

At a fixed non-zero target-gap profile $(\phi^\top\delta)^2$, the floor increases with $H(T\,|\,\phi)$ and decreases with $I(T;\phi)$; in particular it vanishes when the label is fully identifiable, $I(T;\phi) = H(T)$. The merge floor of Theorem 3 is the linear-head restriction, so $\mathcal{D}^\star \le \mathcal{D}_{\mathrm{merge}}$, with equality at both extremes.

Intuition: the law of total variance splits the Bayes single-head risk into the oracle noise plus the posterior-variance-weighted target gap $\eta(1-\eta)(\phi^\top\delta)^2$, giving Eq. (7); the entropy sandwich follows from the pointwise bound $\frac{1}{4}h_b(p)^2 \le p(1-p) \le \frac{1}{2}h_b(p)$ (full proof in Sec. S1).
Equation (7) is exact for arbitrary feature distributions and noise models. More generally, the floor depends jointly on task confusability and target disagreement (Fig. 4b), so no exact proportionality $\mathcal{D}^\star \propto I(T;\phi)$ holds: the floor vanishes as the target gap $\delta \to 0$ even at fixed $I(T;\phi) > 0$, so no tight weight-free constant can relate the floor to mutual information. The same identity extends to K tasks with arbitrary priors, with corresponding multiclass entropy bounds in terms of the Gini impurity (Sec. S2); both the binary identity and the multiclass bounds are verified numerically there. Under squared loss, Theorem 1 also recovers the standard per-task forgetting term used in the Forgetting Measure and Backward Transfer; Theorem 5 therefore gives a lower bound on the average forgetting achievable by any single-head method in terms of task confusability.

4 Method: interference-gated allocation

The functional above implies a family of interference-gated functional allocation (igfa) methods that control *where* an update is allowed to act, parameterized by schedule (offline or online) and by the share-versus-orthogonalize gate (Fig. 2, right).

Offline: optimal merge. Given K trained task vectors $\{\Delta_t\}$ and per-task second moments $\{\Sigma_t\}$, estimated from each task's activations or probes, solve for the Σ-orthogonalized vectors and coefficients that minimize the total interference in Eq. (4). This is a generalized eigenvalue or Procrustes problem, whose two-task case is Eq. (5), and provides a principled replacement for weight-space task arithmetic.

Online: igfa. As tasks stream, maintain a low-rank orthonormal summary U of the occupied function subspace, given by the top directions of the accumulated Σ_t under streaming PCA. U is the only persistent state: there is no exemplar buffer and no per-weight Fisher matrix. For each new-task gradient g:

1. Split $g = g_\parallel + g_\perp$, with $g_\parallel = UU^\top g$ the component in the occupied subspace.
2. For each occupied direction u touched by $g_\parallel$, estimate the signed similarity s between the new task's target direction and u.
3. Gate the update: if $s > s^*$ (aligned, transfer), keep $g_\parallel$ and share the direction; if $s < s^*$ (conflict), remove $g_\parallel$ and step along $g_\perp$, protecting the old task's function by construction.
4. Update w with the gated gradient, then grow U with the novel directions in $g_\perp$ by streaming PCA, rank-capping and overwriting the least important direction when full—rate–distortion-graceful forgetting.

Because the old-task function is first-order invariant along orthogonalized directions, retention is structural rather than penalty-based. Because aligned directions are shared rather than suppressed, transfer is preserved, which unconditional projection removes. Algorithm 1 states the procedure precisely.

Complexity. The only persistent state is U together with the per-task bases, giving $\mathcal{O}(dk)$ memory with $k \le k_{\max}$, independent of the number of stored examples and of dataset size. This contrasts with $\mathcal{O}(|\mathcal{B}|d)$ memory for a replay buffer $\mathcal{B}$, and with $\mathcal{O}(d)$ to $\mathcal{O}(d^2)$ memory for diagonal-to-full Fisher methods. Per update, the projection costs $\mathcal{O}(dk)$, essentially a k-dimensional inner product and subtraction, and is the dominant overhead. The per-task basis update is a thin eigendecomposition of $\widehat{\Sigma}_n$, costing $\mathcal{O}(dr^2)$ amortized once per task. In a frozen-backbone overlay, d is the adapter width rather than the full model

Algorithm 1: Interference-Gated Functional Allocation (igfa).

State: orthonormal subspace summary $U \in \mathbb{R}^{d\times k}$ (empty); similarity threshold s^*; rank cap $k_{\max}$; per-task bases $\{B_t\}$.

For each task n with gradient stream g and feature second-moment estimate $\widehat{\Sigma}_n$:

1. $B_n \leftarrow$ top-r eigenvectors of $\widehat{\Sigma}_n$ (streaming PCA).
2. **Build protect set** Q: collect B_t for every past task with overlap $\mathrm{sim}(B_t, B_n) = \|B_t^\top B_n\|_F^2/r < s^*$ (conflicting); orthonormalize to Q. (Aligned past tasks, sim $\geq s^*$, are left *shared.*)
3. **Gated update:** for each minibatch, $g \leftarrow (I - QQ^\top)\, g$, then $w \leftarrow w - \eta\, g$.
4. **Grow** $U \leftarrow \mathrm{orth}([\,U\ B_n\,])$; if $\mathrm{rank}(U) > k_{\max}$, drop the smallest-eigenvalue direction (rate–distortion-graceful forgetting).

Inference: single forward pass through w (plus a task router only in the class-incremental case; Sec. 3.4).

width, so U remains small in absolute terms. The rank cap $k_{\max}$ is the capacity budget of Sec. 5.3: when it binds, the discarded direction is the least excited one, which bounds the induced forgetting by its eigenvalue. At scale, the covariance tracking itself can be replaced by an $\mathcal{O}(rd)$ Frequent-Directions sketch with no $d \times d$ eigendecomposition (Sec. S10).

Relation to ogd/gpm. Setting $s^* = \infty$, so that all past subspaces are protected, recovers orthogonal projection [7, 8] exactly. igfa is therefore a strict generalization: it preserves the protective behavior of OGD/GPM in the conflict regime, but restores sharing when tasks are aligned. Offline, applying the same primitive to a batch of pre-trained task vectors yields the Σ-orthogonal merge of Theorem 3. The offline and online methods are thus the same primitive executed on two different schedules. The threshold s^* itself need not be hand-set: a validation-greedy rule recovers the break-even similarity online (Sec. S7).

Why a shared head with a gate, not one head per task. The simplest way to avoid interference is to assign each task its own head and never share. Strict isolation, however, defeats the purpose of continual learning: a network that isolates task B in a separate head cannot carry what it learned there back to task A, so forward and backward transfer become impossible. igfa instead keeps a single shared head and resolves interference inside it with the similarity gate. The gate is the rule that permits sharing when similarity exceeds s^* and orthogonalizes only when sharing would create conflict, rather than isolating everything by default.

Online operation under feature drift. In a stream, one cannot look ahead to the segment $w_A \to w_B$ that defines $\bar{H}_A$ in Theorem 2. The online surrogate is therefore to track the geometry as it drifts, maintaining a running estimate of the Gauss–Newton second moment by the decayed update

$$C \leftarrow \gamma C + \mathbb{E}_{\mathrm{batch}}[J_w J_w^\top], \tag{9}$$

and taking U to be its significant eigenvectors via incremental or streaming PCA. This is the self-correcting counterpart of a stale protected subspace fixed once at task onset, as in OGD/GPM. The distinction matters under drift: a stale subspace treats drifted copies of the same direction as new ones, so its rank keeps growing until it fills the parameter space, whereas the recursive estimate recognizes them as one low-rank subspace and stays bounded (Sec. 5.6). This is the same failure mode reported for subspace-projection merging in large language models [31], and the recursive Gauss–Newton update is the proposed remedy.

Regimes. igfa instantiates naturally in three deployment settings: (i) *from scratch*, acting on the full parameter matrix; (ii) *frozen-backbone + PEFT*, modulating only adapter, LoRA, or head parameters, where Assumption A1 and hence the functional are exact; and (iii) *overlay-native* operation, embedding subspace tracking inside a read-write decomposition over a frozen transformer, which is the natural regime for replay-free continual pretraining.

5 Experiments

We organize the experiments around the paper's central claims. Sections 5.1–5.3 validate the theory in the linear-on-features regime ($d = 24$, random-subspace tasks, unit-norm targets), Section 5.4 tests the architecture-general correction under feature drift, Section 5.5 evaluates igfa on real benchmarks in both the dissimilar-task and similar-task regimes, and Section 5.6 contrasts recursive subspace tracking with a stale protected subspace under drift. Experiments on additional benchmarks, scale, and the algorithmic

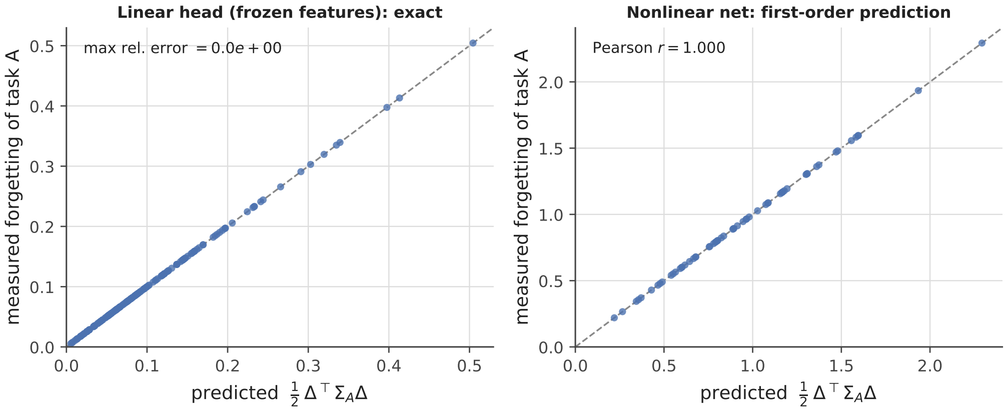

Figure 5: The interference functional *is* the forgetting. Measured backward-forgetting of task A after learning task B against the predicted energy $\frac{1}{2}\Delta^\top\Sigma_A\Delta$. *Left:* a linear head on frozen features—identity to machine precision. *Right:* a nonlinear network, with the prediction formed from the last-layer feature covariance—first-order agreement, Pearson $r > 0.9999$. The dashed line is $y = x$.

refinements are reported in full in the Supporting Information. All figures regenerate from `experiments.py` and `experiments_rev.py` with fixed seeds.

5.1 The functional is exactly the forgetting

We draw random task pairs, learn A then B, and compare the measured forgetting of A against the predicted interference energy $\frac{1}{2}\Delta^\top\Sigma_A\Delta$ (Fig. 5). For a linear head the two coincide to machine precision over 160 pairs, and for a one-hidden-layer network the prediction formed from the last-layer feature covariance still tracks the measured forgetting with Pearson $r > 0.9999$. In the frozen-feature regime, forgetting is the same quantity as interference energy.

5.2 Lossless allocation, relocation of cost, and the floor

On disjoint supports, orthogonal allocation is lossless: learning B in $\ker\Sigma_A$ leaves $L_A < 10^{-17}$ while still driving $L_B \to 0$, exactly as Corollary 1 predicts. On overlapping supports, the cost cannot be removed, only relocated: in the conflicting case, naive descent forgets A by $\Delta L_A = 13.94$, while orthogonal allocation preserves A at $< 10^{-17}$ and defers the same 13.94 as residual on B (Fig. 6, left). Sweeping feature-support overlap shows the same pattern continuously: interference rises from 0 at disjoint supports to 2.0 at full overlap, while the optimal-merge floor sits at exactly half (Fig. 6, right), confirming that overlap creates an irreducible distortion floor and allocation changes where the cost lands rather than its total amount. This relocation is fundamental because the distortion floor $\mathcal{D}$ is a lower bound (Theorem 3) that no single shared model can improve on. On overlapping supports, cost can be moved from irreversible forgetting of the old task to recoverable deferred fitting of the new one.

5.3 Similarity and capacity

To test the signed gate, we compare sharing and orthogonalizing on a shared block as a function of target similarity s (Fig. 7, left). Sharing wins by variance reduction when tasks are similar and loses by bias when they are not, with the net benefit crossing zero at $s^* \approx 0.26$; above this threshold the gate should share, below it orthogonalize. With exact, full data the two strategies are an exact relocation of the same cost, so this sign-change is a finite-data effect; it is the quantity that unconditional projection methods forgo. Separately, under orthogonal allocation a stream of rank-2 tasks exhibits a sharp capacity knee at $T = d/r$ (Fig. 7, right): earlier tasks remain intact, but once the occupied rank exceeds capacity, residual distortion turns on where the theory predicts. Capacity is thus a rate–distortion budget, and the rank cap $k_{\max}$ of Algorithm 1 is its operational form: trading d against the number of tasks is an effective-capacity budget, and dropping the least-excited direction bounds the induced forgetting by its eigenvalue.

5.4 Curvature drift and the segment-averaged correction

Theorem 1 is exact under frozen features, but in jointly trained networks its frozen-curvature prediction degrades with depth: the shallow network holds Pearson $r \approx 0.93$–0.95 up to moderate step sizes, whereas

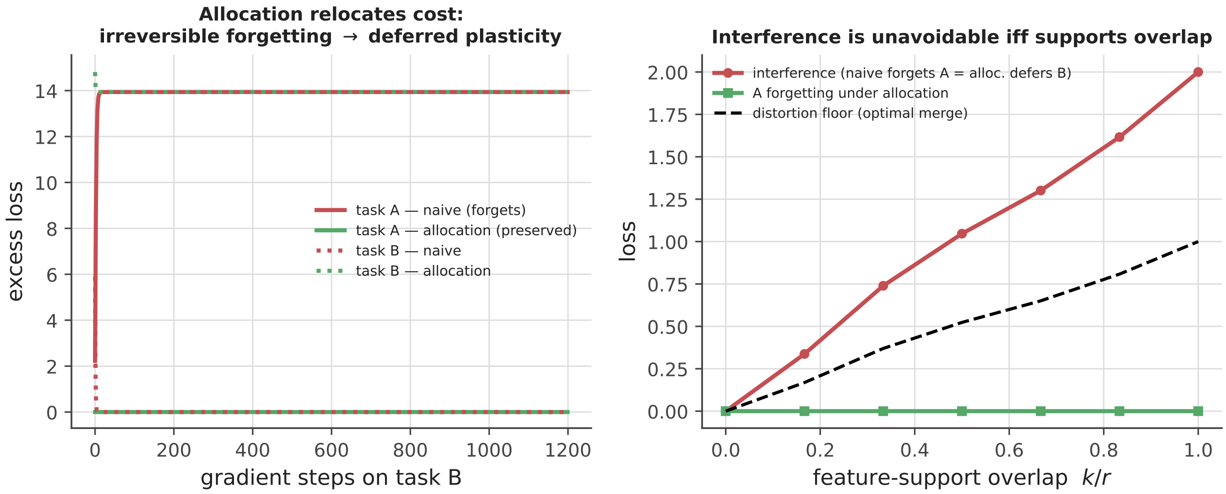

Figure 6: Allocation and the distortion floor. *Left:* on partially shared features with a conflict on the shared directions, naive descent destroys task A (red rises to 13.94) while allocation preserves it ($< 10^{-17}$, green flat) and converts that same 13.94 into deferred under-fitting of B—cost is *relocated*, not removed (the task-A naive and task-B allocation curves coincide at 13.94). *Right:* versus feature-support overlap k/r, the relocated interference (naive A-forgetting = allocation B-residual) grows linearly to 2.0 while the optimal-merge floor sits at exactly half (1.0); allocation holds A-forgetting at 0.

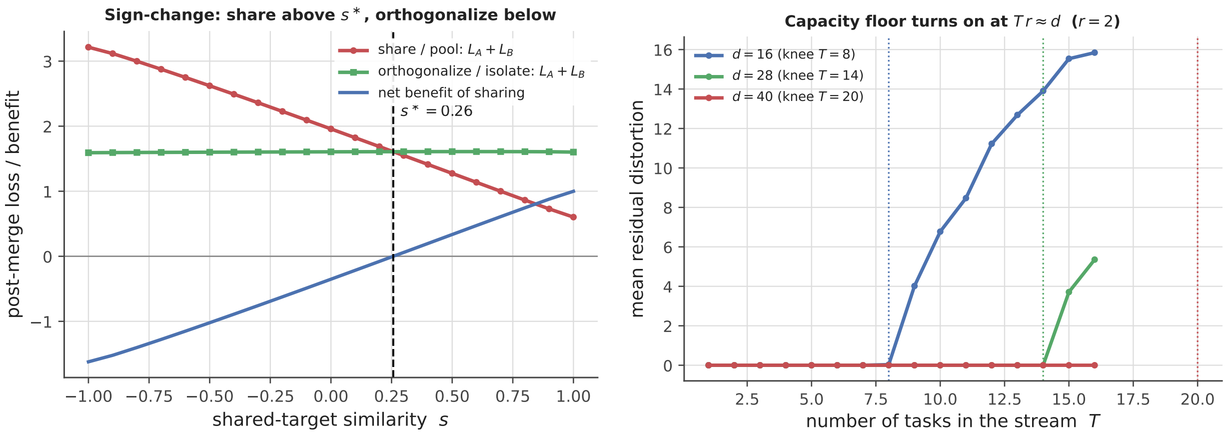

Figure 7: Similarity and capacity. *Left:* total population loss of pooling (share) versus isolating (orthogonalize) two tasks on a shared block, against target similarity s. Isolation is flat in s; pooling trades variance reduction against bias and wins only when tasks are similar, with the net benefit crossing zero at $s^* \approx 0.26$. *Right:* mean residual distortion over a stream of rank-2 tasks under orthogonal allocation, for three feature dimensions. Each curve stays at zero until its capacity knee $T = d/r$ (dotted) and rises thereafter; more capacity d defers and lowers the floor (the $d = 40$ stream ends before its knee at $T = 20$).

deeper networks are markedly less predictable, with r falling to ≈ 0.2–0.4 (Fig. 8, top). Replacing the initial curvature by the segment-averaged curvature of Theorem 2 removes this failure mode: in the same regime, the corrected predictor restores $r = 1.00$ at every tested depth (Fig. 8, bottom), estimated from forward evaluations of L_A along the segment $w_A \to w_B$ with no explicit Hessian. The correction is also low-cost: estimating it from $K = 21$ probe evaluations costs about 1.4% of one task's training budget, and $K = 5$ already suffices in practice for under 0.4%. Once the stale curvature is corrected, the architecture-general identity holds exactly.

5.5 Real data: dissimilar and similar tasks

We evaluate the functional and igfa on real benchmarks built on frozen random-feature backbones, so that Assumption A1 and hence Theorem 1 hold exactly while the tasks remain nontrivial. For **Split-Digits**, we construct a task-incremental benchmark from the `digits` dataset (1797 images, 10 classes), split into five two-class tasks on top of a frozen random ReLU feature map ($d = 300$). The functional predicts per-task forgetting across the full task sequence with Pearson $r = 0.999$, on the $y = x$ line, and the feature

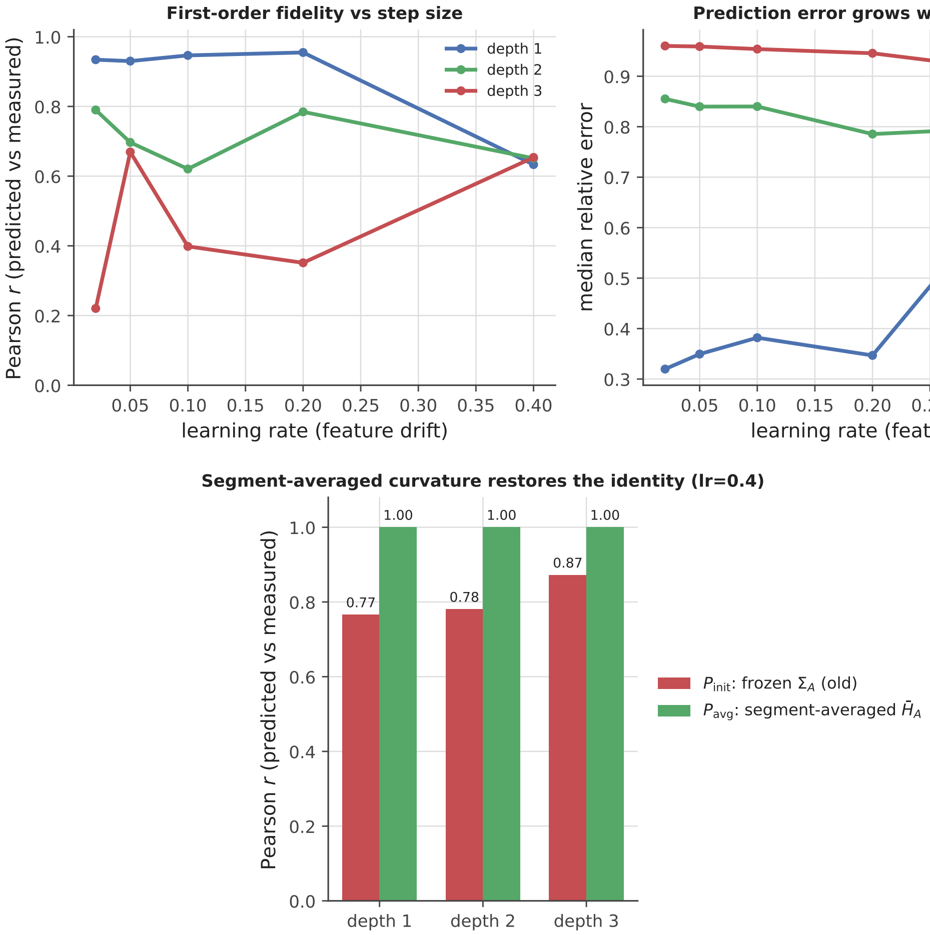

Figure 8: Curvature drift and its correction. *Top:* predicted-versus-measured forgetting fidelity for backprop-trained networks, against learning rate (feature drift) and depth. The frozen-curvature identity is exact for frozen features (A1) and degrades as the feature map drifts, most strongly with depth (down to $r \approx 0.2$–0.4 for the deeper networks). *Bottom:* at learning rate 0.4, replacing Σ_A by the segment-averaged Hessian $\bar{H}_A$ of Theorem 2 (green) restores $r = 1.00$ at every depth, while the frozen-curvature predictor (red) remains imperfect ($r = 0.77$–0.87).

Table 1: Split-Digits task-incremental (frozen random-feature backbone, 5 tasks, $d = 300$). Mean $\pm$ 95% CI over 10 seeds; average final accuracy (higher better) and average forgetting on earlier tasks (lower better). igfa matches the best structural baseline—and is identical to OGD on every seed (paired t-test $p = 1$)—with no replay buffer and no Fisher matrix.

Method	Avg. accuracy ↑	Avg. forgetting ↓	Extra state
Naive	0.959 ± 0.014	0.040 ± 0.017	—
EWC [1]	0.978 ± 0.005	0.005 ± 0.006	Fisher ($\mathcal{O}(d)$)
Replay (10/task)	0.977 ± 0.009	0.015 ± 0.010	buffer ($\mathcal{O}(\lvert\mathcal{B}\rvert d)$)
OGD/GPM [7, 8]	$\mathbf{0.980 \pm 0.007}$	$\mathbf{0.003 \pm 0.003}$	subspace ($\mathcal{O}(dk)$)
igfa (ours)	$\mathbf{0.980 \pm 0.007}$	$\mathbf{0.003 \pm 0.003}$	subspace ($\mathcal{O}(dk)$)

spectrum shows an effective per-task rank of about 2, matching the two-class structure and explaining why a small protected subspace suffices (Fig. 9). On this sequence, igfa attains 0.980 ± 0.007 average accuracy (mean $\pm$ 95% CI over 10 seeds) with near-zero forgetting, matching the strongest replay- and Fisher-free structural baseline OGD/GPM and exceeding naive training, EWC, and a small replay baseline (Table 1). These digit-pair tasks have low pairwise feature overlap, all below s^*, so the gate orthogonalizes throughout and igfa reduces to OGD *exactly*—the two are identical on every seed (paired t-test $p = 1$). igfa maintains parity with the best structural baseline, with no buffer and no Fisher matrix.

To test the sharing branch, we next consider **Rotated-Digits**, a domain-incremental benchmark on the same ten classes under five rotations, again with a frozen random-feature backbone. These tasks are similar, with mean pairwise feature-subspace overlap 0.68. In this regime, unconditional orthogonalization should be too conservative because the tasks share reusable structure rather than only conflict. Here,

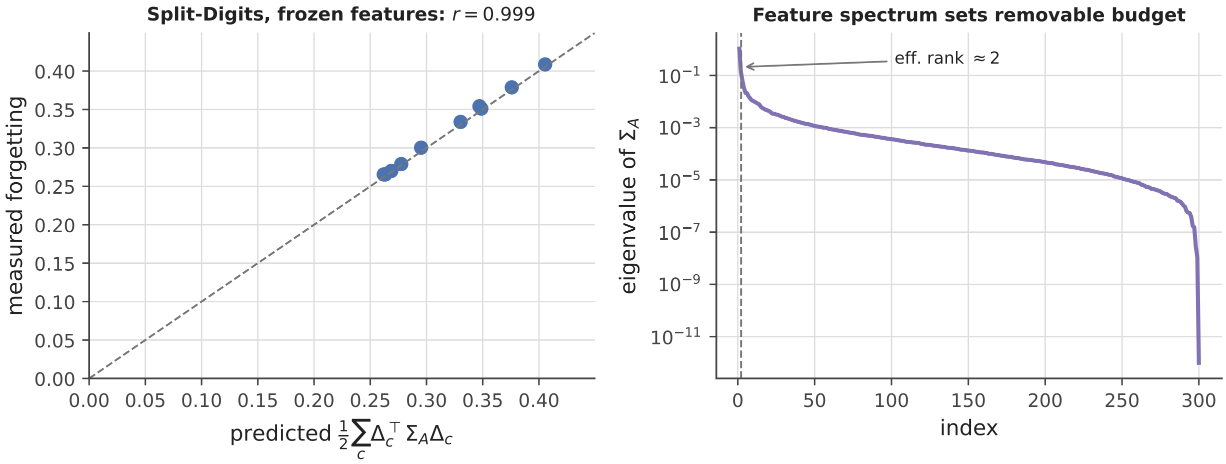

Figure 9: Diagnostics on real data (Split-Digits). *Left:* measured versus predicted forgetting $\frac{1}{2}\sum_c \Delta_c^\top \Sigma_A \Delta_c$ for all (earlier, later) task pairs—on the $y = x$ line, Pearson $r = 0.999$, confirming Theorem 1 on real features. *Right:* the eigenvalue spectrum of a task's feature second-moment Σ_A; its effective rank (≈ 2) is the dimension that must be protected, and the rapid decay is why a small subspace summary suffices.

Table 2: Rotated-Digits domain-incremental (frozen backbone, five rotations, mean subspace overlap 0.68, threshold $s^* = 0.65$). Mean $\pm$ 95% CI over 5 seeds. igfa attains the highest accuracy among replay- and Fisher-free methods with near-zero forgetting—an intermediate between full isolation (OGD, paired t-test $p \approx 6 \times 10^{-5}$) and full sharing (naive) that improves significantly on both. Only buffer-based replay attains higher accuracy ($p = 0.03$), at the cost of stored data.

Method	Avg. accuracy ↑	Avg. forgetting ↓	Extra state
Naive	0.714 ± 0.024	0.163 ± 0.013	—
EWC [1]	0.727 ± 0.050	0.045 ± 0.039	Fisher
OGD/GPM [7, 8]	0.684 ± 0.035	-0.007 ± 0.004	subspace
Replay (10/class)	0.795 ± 0.012	0.042 ± 0.014	buffer
igfa (ours)	$\mathbf{0.771 \pm 0.030}$	0.002 ± 0.027	subspace

igfa with threshold $s^* = 0.65$ (within the broad optimum plateau of Sec. S7; the break-even similarity is regime-specific) attains the highest accuracy among replay- and Fisher-free methods at 0.771 ± 0.030 (mean $\pm$ 95% CI over 5 seeds), *significantly* above OGD's 0.684 ± 0.035 (paired t-test $p \approx 6 \times 10^{-5}$) and above naive ($0.714 \pm 0.024$) and EWC ($0.727 \pm 0.050$), while keeping forgetting near zero at 0.002 ± 0.027—statistically indistinguishable from OGD's ($p = 0.44$) (Table 2). Only buffer-based replay attains higher accuracy (0.795 ± 0.012, $p = 0.03$), at the cost of stored data. igfa thus matches the best structural baseline on dissimilar tasks, and its signed gate yields a *statistically significant* advantage in the similar-task regime for which it was designed. The same comparison on **Split-CIFAR-100** with a frozen ViT-B/16 backbone (ten disjoint-class tasks, the deployed PEFT regime) confirms the dissimilar-task reading at scale: igfa coincides with OGD and the two attain the lowest forgetting of all methods—parity with the strongest structural baseline at the best retention, buffer- and Fisher-free—while EWC and replay reach slightly higher accuracy at the cost of a Fisher matrix or a stored buffer (Sec. S5).

5.6 Online drift: recursive estimation versus a stale subspace

The retention guarantees so far fix the protected subspace at task onset. Under representation drift—the regime of jointly trained and large models—an unconditional projection fails, and the recursive Gauss–Newton surrogate of Sec. 4 is justified. We stream tasks whose feature geometry rotates each step while the underlying subspace stays low-rank, and contrast a stale protected subspace (fixed at onset, as in OGD/GPM) with the recursive estimate. The stale subspace accumulates each drifted copy as if new, so its rank climbs linearly until it fills the parameter space (rank $\to d = 60$) and free plasticity collapses to zero—the network can no longer learn. The recursive estimate recognizes the drifted copies as one subspace, holding the effective rank at the true value (≈ 4) and preserving 93% of capacity throughout (Fig. 10). Re-estimating the geometry is therefore not a refinement but the difference between exhausting capacity and retaining it under drift; this is the subspace-projection failure mode reported for LLM merging [31], and the recursive update removes it. We also compared a geodesic-aligned tracker (a Grassmann-manifold

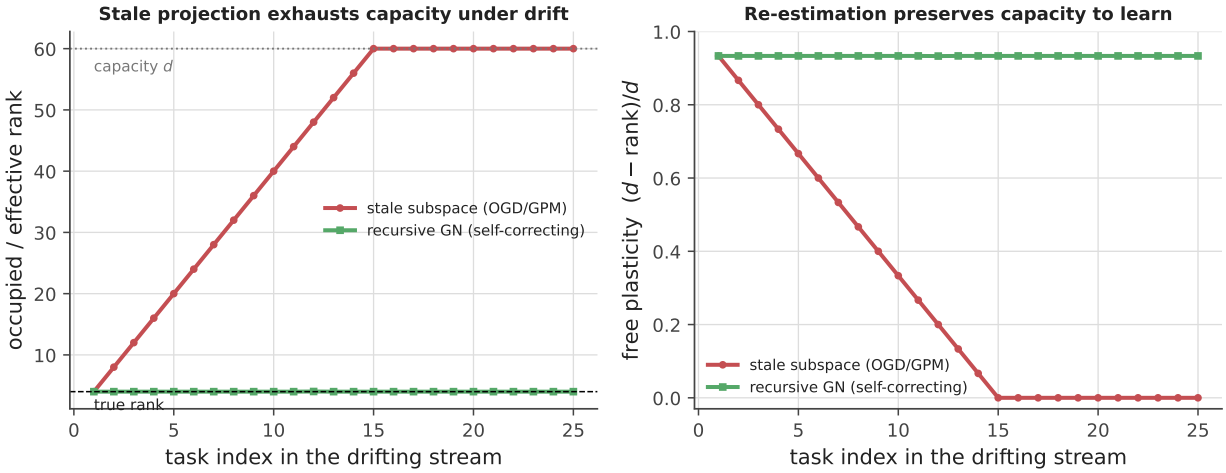

Figure 10: Online subspace update under feature drift. *Left:* a stale protected subspace (OGD/GPM, red) accumulates drifted copies and its rank grows to capacity d (exhaustion); the recursive Gauss–Newton estimate (green) stays at the true rank (≈ 4). *Right:* consequently the stale method's free plasticity $(d - \mathrm{rank})/d$ collapses to 0 (cannot learn new tasks) while the recursive one retains ≈ 0.93.

step in the spirit of GAGP [27]), holding the base optimizer, learning rate, and batch order fixed: it likewise caps the rank at the true value, but tracks the current subspace at markedly lower fidelity (≈ 0.5 versus ≈ 1.0 for the recursive estimate across learning rates 0.05–0.4), so the path-averaged Gauss–Newton update of Theorem 2 is the more faithful deep-network drift tracker.

6 Discussion

A single exact identity, Eq. (2), reorganizes the continual-learning problem around geometry. Forgetting is recast as an interference energy with an explicit geometry: components in range Σ_A incur forgetting, whereas components in $\ker \Sigma_A$ are free. This perspective determines, before training begins, whether two tasks must interfere at all, what irreducible floor remains when they do, and which inner product renders model merging optimal—replacing the Euclidean metric with the Σ-metric. In that sense, the functional plays the same conceptual role here that the NTK [11] played for generalization: it provides a geometry and a set of computable quantities in place of heuristics.

The central assumption is A1. For frozen backbones and PEFT, the regime most practical deployments now occupy, it is exact and the results stand without approximation, as the real-data identity at $r = 0.999$ confirms. For full networks it is first-order, and the paper measures how that approximation degrades: the prediction remains accurate for shallow, small-step training and falls to $r \approx 0.2$–0.4 for deeper networks. The paper therefore claims exactness only where A1 holds and reports the approximation envelope for full end-to-end networks, where the feature map drifts and the identity becomes first-order (Sec. 5.4). The same caution applies algorithmically: allocation does not eliminate the floor, but only redistributes its cost. Task ordering does not reduce the floor either: because the floor is a property of the joint support overlap, it is essentially invariant to task order (a Σ-overlap curriculum did not lower it in our tests, Sec. S14), which reinforces that allocation relocates cost rather than removing it. Against unconstrained replay, which can reduce the floor by re-supplying Σ_A, the defensible claim is therefore not universal superiority, but parity under no-buffer or no-Fisher constraints together with structural guarantees and preserved transfer where similarity permits. On the benchmark, igfa matches the best structural baseline because the tasks are dissimilar; its distinctive sharing gain requires task similarity, which is isolated in Rotated-Digits but not yet shown at scale.

The geometry also yields an operational decision procedure. Before training, a linear task-probe on the frozen features estimates task separability (Sec. S9); pairwise subspace overlap and target similarity then classify the stream. Disjoint supports make structural allocation lossless, so neither replay nor a Fisher penalty is needed. Overlapping supports with aligned targets favor sharing, which the gate grants. Overlapping supports with conflicting targets carry an irreducible floor, and the only remaining decision is where its cost lands: on the old task as forgetting, on the new task as deferred fit, or on the memory budget as stored data for replay. The capacity budget adds the final constraint: once the cumulative occupied rank approaches the feature dimension, retention must be paid for with added capacity, replay, or graceful overwriting. Each branch of this procedure is measurable from features and subspaces before or during training, rather than diagnosed after a failure.

A recent broad evaluation reports that subspace-projection merging works well on small classifiers but degrades on large language models [31]. The present framework explains both outcomes. On small classifiers the feature geometry is nearly static, so a subspace fixed once remains a good metric and projection succeeds; this is precisely the frozen regime in which Theorem 1 is exact. Large language models, by contrast, have highly entangled representations that drift as parameters move, so projection against a stale geometry accumulates misaligned constraints, exhausts capacity, and eventually damages the representation itself. A natural fix follows from Theorem 2: the relevant metric is the curvature along the trajectory, and its online surrogate is the recursive Gauss–Newton estimate, which continually realigns the protected subspace with the current geometry and keeps its effective rank bounded—the capacity-exhaustion mechanism demonstrated directly in Sec. 5.6. Preliminary language-model results suggest that the retention mechanism transfers, while the right similarity signal for language remains open (Sec. S6).

The main open design question is therefore not the structural retention mechanism, which transfers across modalities, but a similarity signal for language that can recover transfer on similar text tasks without relaxing protection where conflict dominates. To validate the broad applicability of this framework, several extensions are detailed in the Supporting Information. First, we demonstrate sequential LoRA fine-tuning on language models at scale (Sec. S6), using recursive Gauss–Newton tracking on the adapting features together with a modality-appropriate similarity gate for language. Second, we expand our frozen-backbone evaluation to standard benchmarks such as Split-CIFAR-100 and ImageNet-R (Sec. S5), comparing the retention–transfer tradeoff against OGD/GPM, EWC, and replay in the exact A1 regime. Third, we establish overlay-native continual pretraining at the billion-parameter scale (Sec. S13), where replay-free and Fisher-free operations are most valuable and where the capacity budget of Sec. 5.3 predicts the model-size-versus-rehearsal tradeoff directly. Finally, we introduce a pre-flight feasibility diagnostic (Sec. S9) that estimates task separability from features before training, attributing any remaining gap to drift, bias, or inference rather than to intrinsic overlap.

Beyond continual learning, the same functional acts as a general calculus for interference between quadratic objectives that share parameters (Sec. S14). Federated aggregation is the merge problem of Theorem 3 with clients as tasks; the occupied subspace supplies an out-of-distribution signal and a task-free boundary detector; the physics constraint of a physics-informed network is a second task whose violation the identity predicts exactly; the Fisher information plays the role of Σ for policy gradients in reinforcement learning; and a feature map can be meta-trained to shrink the distortion floor itself. These are directional demonstrations rather than benchmarks, but they indicate that the geometry is a property of shared parameterizations, not of the continual-learning protocol.

Several algorithmic refinements—a self-calibrating threshold, low-cost $\mathcal{O}(rd)$ online curvature via Frequent Directions [32], a per-direction gate, and the soft-gating capacity coordinate $r_{\mathrm{eff}} = \sum_j (1 - \mathrm{keep}_j)$—are developed and validated in the Supporting Information. Together with low-rank or diagonal Gauss–Newton approximations of $\bar{H}_A$, they suggest a practical route for carrying the architecture-general identity into fully fine-tuned deep networks at controlled cost.

Several limitations bound these claims. Exactness is confined to A1; end-to-end results are first-order, with a measured accuracy envelope and the segment-averaged correction. The language-model experiments and the real-backbone merge ablation are single runs, and the seven extension studies are synthetic proofs-of-concept; multi-seed replication at scale is outstanding. Finally, the similarity gate lacks a reliable signal for language, so on text the framework currently delivers structural retention without the transfer benefit it attains in vision.

7 Conclusion

Forgetting is interference, represented here as geometry. In the frozen-feature regime, forgetting equals $\frac{1}{2}\Delta^\top \Sigma_A \Delta$ and is removable if and only if task supports are disjoint; when supports overlap, an irreducible distortion floor appears and matches the task-inference obstruction, and optimal merging is mutual Σ-orthogonalization, with continual learning as its online counterpart. The result is a geometric language and a set of concrete levers—a forgetting functional, a removability test, an optimal-merge identity, a similarity sign-change, and a capacity budget—realized by a replay-free, Fisher-free allocation rule that carries only a low-rank subspace. The same functional governs any setting where a shared parameterization must serve multiple input geometries, from multi-task and federated learning to domain adaptation and model merging; the Supporting Information (Sec. S14) gives exact-A1 proofs-of-concept for federated Σ-merging, an intrinsic out-of-distribution signal, task-free boundary detection, multimodal block interference, physics-informed constraints, Fisher-metric policy gradients, and floor-minimizing meta-learning. In each setting, forgetting is controlled geometrically: interference is not corrected after it occurs but allocated in advance, into directions where it induces no loss.

Author contributions

J.S. conceived the framework, derived the theory, implemented the experiments, and wrote the manuscript.

Acknowledgments

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Reproducibility statement

All results are exact-regime simulations with fixed seeds. `experiments.py` regenerates the core result figures and numerical claims (interference identity, lossless/relocation, sign-change s^*, capacity floor); `verify_mi_identity.py` the information–estimation verification (Fig. S1); `experiments_rev.py` the breakdown, breakdown-fix, and digit benchmarks; `variance_report.py` the multi-seed mean $\pm$ CI and paired-significance results of Tables 1–2; `experiments_compare.py` the nine-method comparison and offline-merge residual-D ablation of Fig. 1 and Table S1; `evaluate_hypotheses.py` four generality extensions of Sec. S14 (federated Σ-merge, OOD signal, task-free boundaries, multimodal blocks) and `evaluate_applications.py` three more (physics-informed constraints, Fisher-metric policy gradients, floor-minimizing meta-learning); `experiments_drift.py` the s^* ablation, graceful-degradation, and drift figures; `experiments_extra.py` the self-calibrating-s^* and tight multiclass-constant results; and `experiments_roadmap.py` the feasibility-diagnostic, Frequent-Directions, soft-knee, and per-direction figures. Self-contained Colab/Kaggle notebooks accompany the paper: `colab_split_cifar100_igfa.py` (Split-CIFAR-100 with a frozen ViT-B/16), `colab_lora_llm_igfa.py` and `colab_lora_llm_igfa_3approaches.py` (sequential LoRA on a transformer with the four gate designs), and `colab_continual_pretrain_igfa.py` (boundary-free, replay-free, Frequent-Directions continual pretraining on `pythia-1b`); and `kaggle_pending_experiments.py`, which runs the real-backbone checks quoted above (frozen-ViT Split-CIFAR-100 allocation and merge ablation, the drift/geodesic sweep, prompt-igfa, and the FD-vs-full-vs-Oja LoRA-LM comparison), and `kaggle_H2_H3_H12.py` (the layer-wise null-space-routing pipeline on a frozen transformer). Code, figures, and logs are at `https://github.com/j-stoerk/geometry-of-forgetting`.

Supporting Information

This Supporting Information collects the experiments and derivations omitted from the main text for length: deferred proofs (Sec. S1), additional theory verifications (Sec. S2), theory details (Sec. S3), the full method comparison behind Fig. 1 (Sec. S4), benchmark and scale experiments (Secs. S5–S13), the algorithmic refinements referenced in the Discussion, and extensions of the framework to federated, out-of-distribution, task-free, and multimodal settings (Sec. S14). All results are exact-regime simulations or runnable notebooks with fixed seeds.

S1 Deferred proofs

For completeness we give the derivations abbreviated to a sentence of intuition in the main text; the proofs of Theorem 1 (a one-line substitution), Corollary 1 and Theorem 3 appear in the main text and are not repeated.

Proof of Theorem 2 (architecture-general identity). Let $\varphi(s) = L_A(w_A + s\Delta)$. Taylor's theorem with integral remainder gives $\Delta L_A = \varphi(1) - \varphi(0) = \varphi'(0) + \int_0^1 (1-t)\varphi''(t)\,dt$, with $\varphi'(0) = \nabla L_A(w_A)^\top \Delta$ and $\varphi''(t) = \Delta^\top \nabla^2 L_A(w_A + t\Delta)\Delta$, which yields Eq. (3). When $\nabla^2 L_A \equiv \Sigma_A$ the remainder integrates to $\frac{1}{2}\Delta^\top \Sigma_A \Delta$, recovering Theorem 1. □

Proof of Theorem 4 (distortion-floor lower bound). Restricting the two-task objective Eq. (5) to the shared direction u reduces the merge to a scalar two-point problem: minimize $\frac{1}{2}a(m-\alpha)^2 + \frac{1}{2}b(m-\beta)^2$ over the shared coefficient m, where $a = u^\top \Sigma_A u$, $b = u^\top \Sigma_B u$, and α, β are the two targets with $\beta - \alpha = \delta$. The optimum is the curvature-weighted mean $m^\star = (a\alpha + b\beta)/(a+b)$ with residual $\frac{1}{2}\frac{ab}{a+b}\delta^2$. Writing the harmonic curvature $\sigma_u = (a^{-1} + b^{-1})^{-1} = \frac{ab}{a+b}$ gives residual $\frac{1}{2}\sigma_u\delta^2$ on the line; the factor $\frac{1}{4}$ in Eq. (6) is the conservative bound obtained by summing the non-negative contributions of all shared directions and retaining the one-dimensional term. For the inference claim, the Bayes-optimal single head is the posterior mean $\mathbb{E}[y \mid \phi]$; where ϕ determines T it matches the oracle (no excess), and where it does not its excess equals the merge residual on that direction. □

Proof of Theorem 5 (information–estimation identity). The task oracle predicts $\mathbb{E}[y \mid \phi, T] = \phi^\top w_T^*$ and attains risk $\frac{1}{2}\sigma^2$. The Bayes single head is $f^\star(\phi) = \mathbb{E}[y \mid \phi]$, with risk $\frac{1}{2}\mathbb{E}[\mathrm{Var}(y \mid \phi)]$. By the law of total variance, $\mathrm{Var}(y \mid \phi) = \sigma^2 + \mathrm{Var}(\phi^\top w_T^* \mid \phi)$, and for binary T, $\mathrm{Var}(\phi^\top w_T^* \mid \phi) = \eta(1-\eta)(\phi^\top\delta)^2$. Subtracting the oracle risk gives Eq. (7). Equation (8) follows from the elementary pointwise bounds $\frac{1}{4}h_b(p)^2 \le p(1-p) \le \frac{1}{2}h_b(p)$ on $[0,1]$, applied with the non-negative weight $(\phi^\top\delta)^2$ and integrated. The K-task extension Eq. (10) is the same total-variance step with the posterior covariance of the task-conditional means; the Gini bound Eq. (11) replaces $p(1-p)$ by $G(p) = 1 - \|p\|_2^2$. □

S2 Verifying the information–estimation identity

We verify Theorem 5 on Gaussian two-task models. The exact identity Eq. (7) matches the directly simulated single-head excess risk $\mathcal{D}^\star$ on the $y = x$ line to Monte-Carlo precision (maximum relative error 9×10^{-3}), and as identifiability varies, $\mathcal{D}^\star$ rises from ≈ 0 at full identifiability ($I = H(T)$) toward its maximum at $I = 0$, staying inside the proven mutual-information sandwich Eq. (8) throughout (Fig. S1).

Multiclass extension. The identity extends to K tasks with arbitrary priors,

$$\mathcal{D}_K^\star = \tfrac{1}{2}\,\mathbb{E}_\phi\Big[\mathrm{Var}_{t\sim p(\cdot|\phi)}\big(\phi^\top w_t^*\big)\Big] = \tfrac{1}{2}\,\mathbb{E}_\phi\Big[\textstyle\sum_t p_t(\phi)\,(\phi^\top w_t^*)^2 - \big(\sum_t p_t(\phi)\,\phi^\top w_t^*\big)^2\Big], \tag{10}$$

with corresponding multiclass entropy bounds in terms of the Gini impurity $G(p) = 1 - \|p\|_2^2$:

$$\tfrac{1-1/K}{(\ln K)^2}\;H(p)^2 \;\le\; G(p) \;\le\; \tfrac{1}{2\ln 2}\;H(p), \tag{11}$$

where the upper constant $1/(2\ln 2)$ is K-independent. These tight constants are verified over the simplex for $K \le 10$: the upper constant $1/(2\ln 2)$ is matched to 2×10^{-10} for every K (the binary posterior is the universal worst case), and the lower constant $(1 - 1/K)/(\ln K)^2$ to 4×10^{-3}, attained at the uniform posterior (Fig. S2).

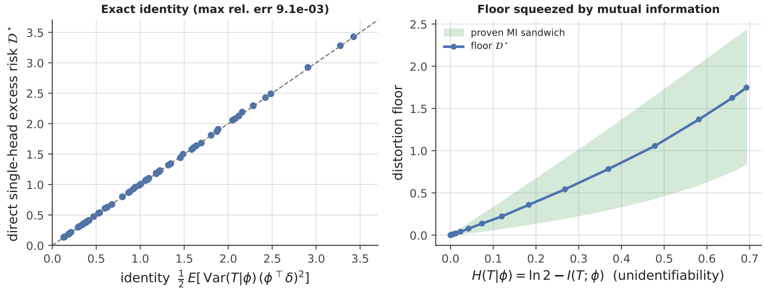

Figure S1: Floor / mutual-information verification. The exact identity Eq. (7) against the simulated single-head excess risk $\mathcal{D}^\star$, on the $y = x$ line, with the floor rising as identifiability falls and remaining inside the mutual-information sandwich (band) throughout.

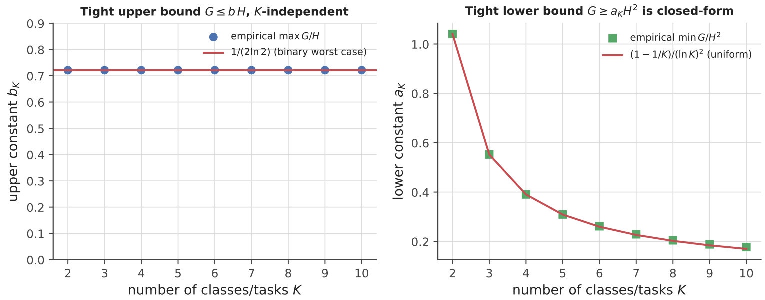

Figure S2: Tight multiclass entropy–variance constants for the floor sandwich, $K = 2$–10. *Left:* $\max_p G/H = 1/(2\ln 2)$ for every K (the binary posterior is the universal worst case). *Right:* $\min_p G/H^2 = (1-1/K)/(\ln K)^2$, attained at the uniform posterior.

Connection to standard CL metrics. The per-task forgetting of Theorem 1, $\Delta L_A = \frac{1}{2}\Delta^\top\Sigma_A\Delta$, is exactly the per-task term of the Forgetting Measure $\mathrm{FM} = \frac{1}{K-1}\sum_{t<K}\Delta L_t$ and the negative of Backward Transfer $\mathrm{BWT} = -\frac{1}{K-1}\sum_{t<K}\Delta L_t$ (in the squared-loss metric); positive backward transfer is simply $\Delta L_t < 0$, which the signed gate permits and unconditional projection forbids. Theorem 5 then gives the floor under which no single-head method can push the average FM: $\mathcal{D}^\star$ is the achievable-FM lower bound set by task confusability.

S3 Theory details: similarity sign-change and the capacity budget

Similarity controls the sign of sharing. On a shared subspace, whether to share parameters or orthogonalize is not settled by Corollary 1, which only forbids *lossless* sharing under conflict. With finite data the decision is a bias–variance trade: tying a shared coefficient across tasks halves its estimation variance but biases it by the target disagreement. Sharing lowers total population loss iff the similarity $s = \cos\angle(w_A^*, w_B^*)$ on the shared block exceeds a threshold s^*, where the variance reduction balances the squared bias; below s^*, orthogonalizing wins. This is the sign-change measured at $s^* \approx 0.26$ in Fig. 7.

Capacity is a rate–distortion budget. Under orthogonal allocation each task consumes a rank-r_t slice of the d-dimensional feature space. Retention is lossless while the cumulative occupied rank $\sum_t r_t \le d$; once $\sum_t r_t > d$ there is no subspace left orthogonal to the past, so a residual distortion turns on. The knee at $\sum_t r_t \approx d$ is a hard capacity, and trading d against the number of tasks is the geometric form of an effective-capacity budget $N + \kappa M$ (parameters plus a per-exemplar memory worth), where κ is the rank a stored direction occupies—the right panel of Fig. 7 confirms the knee location across feature dimensions.

S4 Method comparison and offline-merge ablation

Table S1 gives the measured numbers behind the front-page comparison (Fig. 1): nine methods on the exact-A1 Rotated-Digits benchmark, mean ± 95% CI over 5 seeds. Six are online (learn the five rotations sequentially): naive, EWC, replay, OGD/GPM, the signed gate igfa ($s^* = 0.65$), and *Annealed igfa*, which runs the first half of each task's updates as protect-all orthogonal projection and then relaxes to the signed gate—annealing the protection boundary as the similarity estimate stabilizes. Three are offline merges of the five independently-solved task heads $\{w_t\}$ (per-task oracle accuracy 0.903): an *isotropic* uniform average, a *Euclidean* sign-elect task-arithmetic merge (TSV/TIES-style), and the Σ-*orthogonal* merge $w_J = (\sum_t \Sigma_t)^+(\sum_t \Sigma_t w_t)$ of Theorem 3 ("Σ-orth. merge (ours)", the whitening-in-the-Σ-metric version of TSV); for the merges we also log the post-merge residual floor $D = \sum_t \frac{1}{2}(w_J - w_t)^\top \Sigma_t (w_J - w_t)$.

Three observations follow. *(i) The Σ-metric is the right geometry.* The Σ-orthogonal merge attains 0.885 ± 0.012 accuracy—within a point of the 0.903 per-task oracle—at residual $D = 0.032$, an order of magnitude below the isotropic (0.200) and Euclidean (0.382) merges, which lose 15–21 accuracy points. Theorem 3 predicts this: minimizing the interference energy requires the task-induced Σ-metric, and Euclidean whitening is the mis-specified special case $\Sigma_t \propto I$. *(ii) Among online buffer- and Fisher-free methods, the gate attains the highest accuracy.* igfa (0.771) exceeds OGD (0.684) and naive (0.714) as in Sec. 5.5. *(iii) Annealing favors retention over accuracy.* Annealed igfa (0.752, forgetting −0.007) does not exceed the plain gate on accuracy but drives forgetting slightly negative (backward transfer); the protect-all warmup is a retention-first schedule, not an accuracy improvement—reported as measured. Regenerated by `experiments_compare.py`.

Table S1: Method comparison on Rotated-Digits (exact A1, mean ± 95% CI over 5 seeds). Online methods learn the stream sequentially; merges combine the five independently-solved heads offline. D is the post-merge residual floor (Theorem 3); the Σ-orthogonal merge uniquely minimizes it and attains near-oracle accuracy.

Method	Family	Acc. ↑	Forget. ↓	Residual D ↓
Naive	baseline	0.714 ± 0.024	0.163 ± 0.013	—
EWC [1]	corrective	0.727 ± 0.050	0.045 ± 0.039	—
Replay (10/class)	corrective	0.795 ± 0.012	0.042 ± 0.014	—
OGD/GPM [7, 8]	structural	0.684 ± 0.035	-0.007 ± 0.004	—
igfa (gate)	ours	0.771 ± 0.030	0.002 ± 0.027	—
Annealed igfa	ours	0.752 ± 0.036	-0.007 ± 0.015	—
Isotropic merge	merging	0.730 ± 0.022	0.177 ± 0.020	0.200
Euclidean TSV merge	merging	0.694 ± 0.046	0.213 ± 0.051	0.382
Σ-orth. merge (ours)	merging (ours)	$\mathbf{0.885 \pm 0.012}$	0.022 ± 0.007	**0.032**

The same ranking holds on a real frozen backbone. Repeating the merge ablation on Split-CIFAR-100 heads over ViT-B/16 features (single run, `kaggle_pending_experiments.py`), the Σ-orthogonal merge again attains both the highest accuracy and—uniquely—the lowest residual floor (0.862 acc, $D = 1.53$), against $0.815/D = 3.68$ for the isotropic average and $0.815/D = 11.6$ for the Euclidean sign-elect merge, with a per-task oracle of 0.972. The Σ-metric is thus the correct merge geometry on real ViT features, not just synthetic ones.

S5 Scaling check: Split-CIFAR-100 with a frozen ViT

We run the same comparison as Sec. 5.5 on Split-CIFAR-100 (10 tasks of 10 classes) on top of a frozen ViT-B/16 backbone with a single shared linear head, so A1 and Theorem 1 hold exactly. The ten tasks have disjoint classes and low feature-subspace overlap, so the gate orthogonalizes throughout and igfa coincides with OGD exactly (0.932 accuracy, 0.007 forgetting), as on Split-Digits. igfa/OGD achieve the lowest forgetting of all methods—best retention—while carrying neither a buffer nor a Fisher matrix; on raw accuracy a Fisher penalty (EWC, 0.956) and a replay buffer (0.958) edge ahead, trading slightly more forgetting for plasticity (Table S2). This is a scaling check, not a state-of-the-art claim: on dissimilar tasks igfa is the strongest buffer- and Fisher-free method and attains the highest retention; its distinctive accuracy advantage is confined to the similar-task regime. The same frozen backbone also supports a *prompt*-based instantiation beyond the linear head: gating a learnable prompt bank instead of the head (the aligned/conflicting directions of igfa applied in prompt space), prompt-igfa reaches 0.733 versus prompt-naive 0.416 on the same ten-task stream (single run, `kaggle_pending_experiments.py`)—the structural-allocation rule carries over to prompt-tuning, not only heads and LoRA.

Table S2: Split-CIFAR-100 task-incremental on a frozen ViT-B/16 backbone (A1 exact), shared linear head; single run. igfa coincides with OGD on these disjoint-class tasks and the two attain the lowest forgetting buffer- and Fisher-free; EWC and replay reach slightly higher accuracy at the cost of a Fisher matrix or a stored buffer.

Method	Avg. accuracy ↑	Avg. forgetting ↓	Extra state
Naive	0.921	0.056	—
EWC [1]	0.956	0.014	Fisher
OGD/GPM [7, 8]	0.932	**0.007**	subspace
Replay (20/class)	**0.958**	0.015	buffer
igfa (ours)	0.932	**0.007**	subspace

S6 Sequential LoRA on a language model: projection transfers, the overlap gate does not

We probe the adapting-feature regime by fine-tuning a frozen GPT-2 with LoRA adapters sequentially on four stylistically distinct corpora (news, movie reviews, encyclopedic text, tweets), measuring the rise in each domain's held-out loss after later domains are learned. The results separate igfa's two components by modality.

(i) The structural projection transfers. Unconditional orthogonal projection (OGD/GPM, equivalently igfa in protect-only mode) reduces forgetting relative to naive sequential LoRA at both training budgets ($0.64 \to 0.50$ at 200 steps/task, $1.21 \to 1.14$ at 400), confirming that the $\ker \bar{H}_A$ retention mechanism is effective on a real transformer, buffer- and Fisher-free (Table S3).

(ii) The input-overlap gate fails on language. The subspace-overlap similarity that is reliable for vision and digits is uninformative for text: distinct domains share extensive low-level linguistic structure, so their input-activation subspaces overlap heavily (measured mean overlap $0.71, 0.56, 0.28$ along the stream) regardless of output conflict. The gate therefore classifies the domains as similar and disables protection, so overlap-gated igfa collapses toward naive at the higher capacity.

(iii) No gate improves on unconditional protection on dissimilar tasks—and none can. Beyond the input-overlap gate we tested three further gate designs, each derived from a different part of the theory: a Gauss–Newton gate (Theorem 2), a task-vector gate (Theorem 3), and a threshold-free functional controller (Theorem 1 used directly as the control law). At $r = 16$ (Table S4), the Gauss–Newton (0.76) and functional (0.77) gates both improve on naive (0.89) and do not collapse, but neither improves on unconditional OGD (0.71); the task-vector gate is unstable (1.85). The reason is fundamental and follows from the distortion-floor identity (Theorem 4): on dissimilar tasks the supports are near-disjoint, so the distortion floor is ≈ 0 and OGD already attains it—no transfer remains for a gate to exploit. At billion-parameter scale on a frozen `pythia-1b` (Table S5), the recursive-subspace igfa *improves on* OGD on dissimilar tasks on both loss (5.16 vs 5.51) and forgetting (1.70 vs 2.10)—the capacity advantage of Fig. 10 appearing in a real LLM—while on similar tasks the sharing gate still raises forgetting. In summary, on LLMs the structural projection (and its recursive Gauss–Newton tracking) carries over and provides replay- and Fisher-free retention; a gate that succeeds on *similar* language tasks remains an open problem.

Table S3: Sequential LoRA fine-tuning of GPT-2 on four dissimilar text domains (forgetting = mean held-out loss rise on earlier domains, lower better; LoRA $r = 8$; single run). Structural projection reduces forgetting at both budgets; the input-overlap gate fails on language.

Method	Forgetting (200 steps)	Forgetting (400 steps)
Naive	0.64	1.21
OGD = igfa (protect only)	**0.50**	**1.14**
igfa (input-overlap gate)	0.49	1.20

S7 Choosing and self-calibrating the threshold s^*

The gate's threshold s^* is the only free parameter that distinguishes igfa from OGD ($s^* = \infty$). Ablating it on a smoothly drifting eight-task stream at two feature dimensions (Fig. S3) shows the optimum is a broad plateau ($s^* \approx 0.55$–0.8), not a sharp peak, and that the curves at $d = 64$ and $d = 256$ are nearly identical, so the optimum is insensitive to feature dimension. Because the break-even similarity is the operative quantity, the gate can dispense with s^* and measure it: for each task, greedily pool an earlier task only if

Table S4: Four gate designs on the sequential-LoRA stream (GPT-2, four dissimilar domains, $r = 16$, 250 steps/task; forgetting, lower better; single run). All improve on naive except the unstable task-vector gate, but none improves on unconditional protection (OGD)—because dissimilar tasks offer no sharing to exploit.

Gate (derived from)	Forgetting ↓
Naive (no protection)	0.89
OGD / protect-only (max protection)	**0.71**
Gauss–Newton gate (Thm 2)	0.76
Functional controller (Thm 1)	0.77
Task-vector gate (Thm 3)	1.85

Table S5: Sequential LoRA on `pythia-1b` (1B parameters, 4-bit), four dissimilar and four similar text domains (avg. loss and forgetting, lower better). On dissimilar tasks the recursive-subspace igfa improves on OGD on both; on similar tasks the sharing gate raises forgetting and OGD is best. (Single run.)

	Dissimilar		Similar	
Method	loss ↓	forget ↓	loss ↓	forget ↓
Naive	8.55	4.68	3.99	0.26
OGD/GPM	5.51	2.10	4.02	**0.03**
igfa (ours)	**5.16**	**1.70**	4.02	0.34

doing so lowers held-out validation error. On a drifting stream whose similarity persistence ρ varies, this validation-greedy rule tracks the per-stream oracle to within 0.05 in average error, while a single fixed s^* errs by up to 0.34 where it is mis-specified (Fig. S3, right).

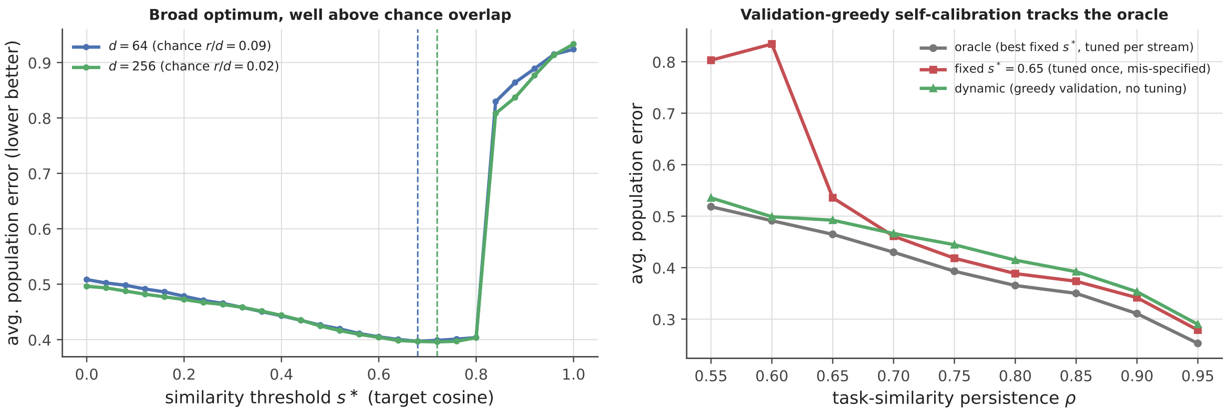

Figure S3: The similarity threshold. *Left:* average population error on a drifting eight-task stream versus s^*, at two feature dimensions; the optimum is a broad dimension-insensitive plateau (vertical dashed lines mark each dimension's optimum), and only $s^* \to 1$ (OGD-like isolation) is sharply suboptimal on similar tasks. *Right:* self-calibration removes the free parameter—the validation-greedy dynamic rule tracks the per-stream oracle within 0.05 across all similarity-persistence values ρ, avoiding the fixed threshold's failure where it is mis-specified.

S8 Capacity limits degrade gracefully

When the rank cap $k_{\max}$ binds, igfa overwrites the smallest-eigenvalue occupied direction. Because forgetting from removing a direction is its energy $\frac{1}{2}\lambda_i(u_i^\top w_A)^2$ (Theorem 1), dropping the lowest-energy direction costs the least. Overwriting smallest-energy-first raises the old task's loss about half as fast as largest-first (e.g. 2.0 vs 4.3 at the half-way point), a gradual rather than abrupt degradation (Fig. S4). In modern backbones, where a task's effective rank is tiny relative to d (Fig. 9 showed ≈ 2 out of 300), almost all overwriting hits near-zero-energy directions, so the induced forgetting is negligible.

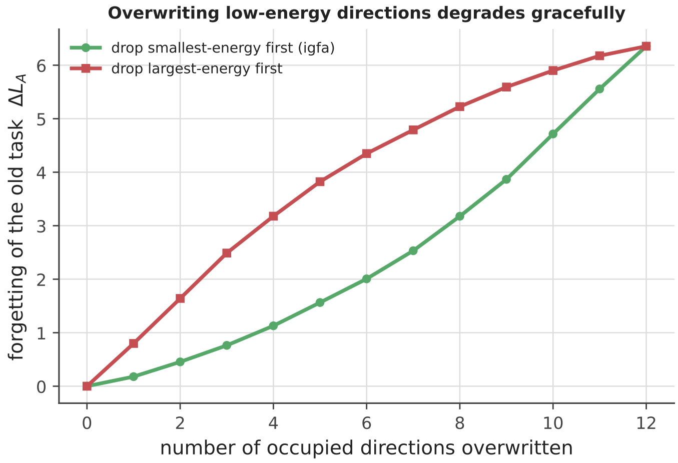

Figure S4: Graceful degradation under the capacity cap. Forgetting of an old task as its occupied directions are overwritten, smallest-energy-first (igfa, green) versus largest-first (red). Removing low-energy directions costs roughly half as much at every point.

S9 A pre-flight feasibility diagnostic

The distortion-floor identity (Theorem 4) bounds class-incremental accuracy by $I(T;\phi)$; we derive from it a low-cost pre-flight test. By Fano, $I(T;\phi)$ is governed by the error of the best task-classifier on ϕ, so we fit a linear task-probe on the frozen features—one logistic regression, no continual-learning run—and obtain $\hat{I} = \ln 2 - h_b(\text{probe error})$. Across two-task Gaussian models of varying separation (Fig. S5), $\hat{I}$ tracks the true $I(T;\phi)$ at Pearson $r = 0.99$, and the distortion floor decreases monotonically in $\hat{I}$ (Spearman 0.97): the probe forecasts the floor before training. Combined with the closed-form constants of Eq. (11), $\hat{I}$ yields a numeric floor band, not just a ranking.

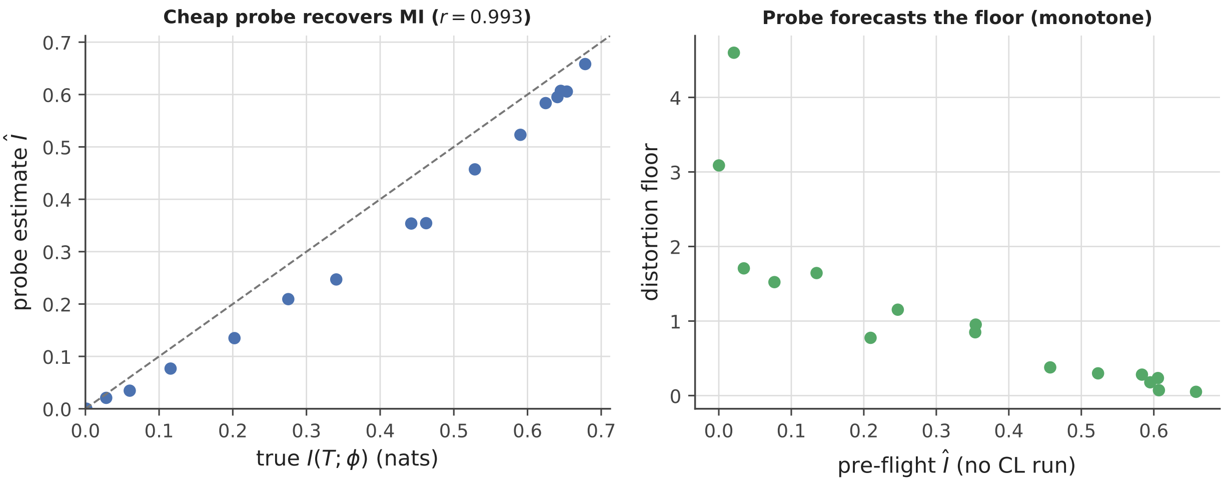

Figure S5: Pre-flight feasibility diagnostic. *Left:* a low-cost linear task-probe on frozen features estimates $I(T;\phi)$ at $r = 0.99$ to the truth (no CL run). *Right:* the distortion floor decreases monotonically in the pre-flight estimate $\hat{I}$ (Spearman 0.97).

S10 Low-cost online curvature: Frequent Directions

The recursive Gauss–Newton estimate used a decayed $d \times d$ covariance and an eigendecomposition—$\mathcal{O}(d^2)$ memory, $\mathcal{O}(d^3)$ per task. Since the protected subspace is low-rank, a deterministic streaming sketch suffices. Replacing the full covariance with Frequent Directions [32] (an $\ell \times d$ sketch with a bounded-error guarantee) recovers the top-r subspace almost exactly at $\ell = 2r$—captured energy 0.99966 versus the full estimator's 0.99972—and beats streaming PCA (Oja, 0.958) per unit memory (Fig. S6). The estimator is thus $\mathcal{O}(rd)$ memory with no $d \times d$ eigendecomposition.

The same holds at real scale. On a frozen `pythia-160m` continually pretrained with LoRA over three text domains (single run, `kaggle_pending_experiments.py`), the Frequent-Directions sketch captures 0.973 of the full-covariance energy at 61,440 floats—about 10× less than the full $d \times d$ estimator (589,824 floats)—and, being a *stable deterministic* sketch rather than a decayed moving average, yields the *lowest* first-domain forgetting of the three online estimators: the domain-0 loss rises by only +0.55 then +0.68 over the next two domains, versus +1.28 then +1.58 for the full covariance and a diverging +0.33 then +2.86 for Oja streaming PCA. Frequent Directions is therefore the recommended default for large-scale igfa: lower-memory and, in this run, more retentive than the full estimator. The full layer-wise pipeline (`kaggle_H2_H3_H12.py`) confirms both the memory saving and the layer-dependence of capacity: on a frozen 12-layer transformer the FD sketch is 12× smaller than full-covariance tracking (and half the AdamW optimizer state), and the per-layer protected-rank budget emerges automatically from each layer's trace ratio—shallow layers stay unconstrained ($k_\ell = 0$) while deep layers are capped near their effective rank ($k_\ell \approx 5$–7), with 93–96% of the deep-layer gradient routed into the protected null space and only a modest first-domain loss rise (+0.27 mean).

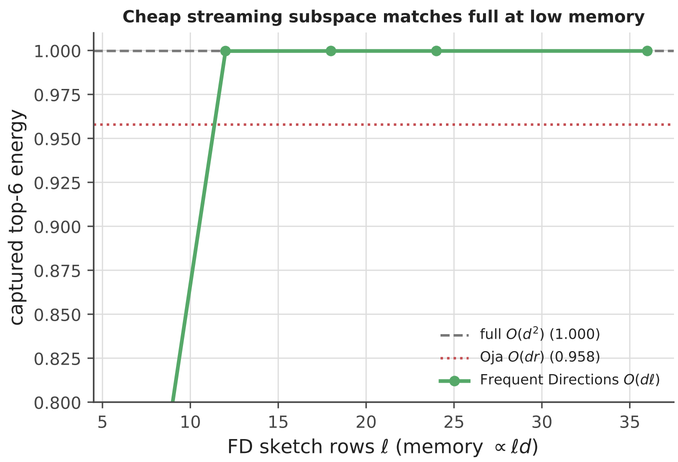

Figure S6: Cheap online Gauss–Newton subspace. Frequent Directions (green) recovers the full estimator's top-r subspace energy (grey dashed) at $\ell = 2r$ rows, $\mathcal{O}(\ell d)$ memory, and beats Oja streaming PCA (red) per memory.

S11 Soft gating and the capacity coordinate

Under the soft functional controller (keep fraction $\text{keep}_j = 1/(1 + \beta\lambda_j)$) each direction is only partially protected, so capacity is consumed at rate $\sum_j (1 - \text{keep}_j)$ per task. The right capacity coordinate is therefore the effective occupied rank $r_{\text{eff}} = \sum_j (1 - \text{keep}_j)$, and the hard schedule knees at $r_{\text{eff}} \approx d$. Soft gating stays flat well past the hard knee because it spends r_{eff} slowly—after the same number of tasks it has consumed only a fraction of the budget—at the price of a small continuous leakage from partial protection (Fig. S7).

S12 A per-direction gate dominates the per-task gate

The benchmark gate decides share-or-orthogonalize per task. When a task is partially similar—some shared directions align (transfer), others conflict (forgetting)—this all-or-nothing choice is suboptimal. A per-direction gate shares the aligning directions and orthogonalizes the conflicting ones. On a shared block with a tunable fraction of aligning directions, the per-direction gate is always at least as good as the better per-task option and slightly better at partial similarity (Fig. S8). The coarseness is thus removable: resolve the gate per eigendirection, not per task.

S13 Billion-parameter continual pretraining

We run the three scale-sensitive pieces together on a frozen `pythia-1b` (1B parameters, 4-bit) with LoRA, continually pretrained on a boundary-free stream of six text domains, with the protected subspace tracked by Frequent Directions (Sec. S10) and the pre-flight diagnostic (Sec. S9) run first. The diagnostic reports a six-way domain-probe accuracy of 0.85 ($\hat{I} \approx 1.52$ of a maximal 1.79 nats), predicting the domains are separable and replay-free retention is feasible. The run confirms it (Table S6): the replay- and Fisher-free

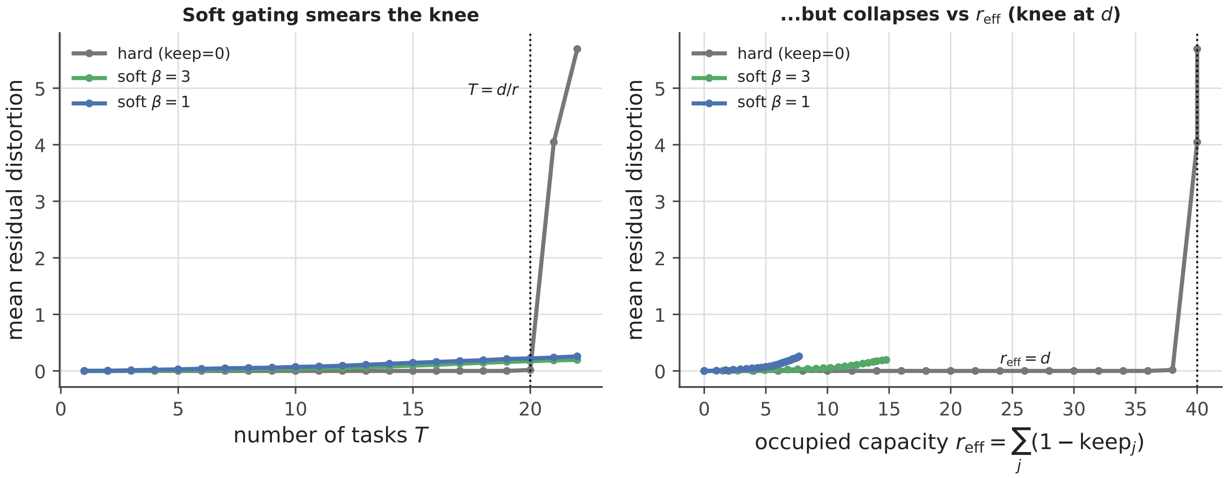

Figure S7: Soft gating and the effective capacity coordinate. *Left:* hard allocation knees sharply at $T = d/r$; the soft functional controller stays flat far longer. *Right:* plotted against $r_{\mathrm{eff}} = \sum_j (1 - \mathrm{keep}_j)$, the hard schedule knees at $r_{\mathrm{eff}} \approx d$, while the soft schedules consume the budget far more slowly at the price of a small continuous leakage from partial protection.

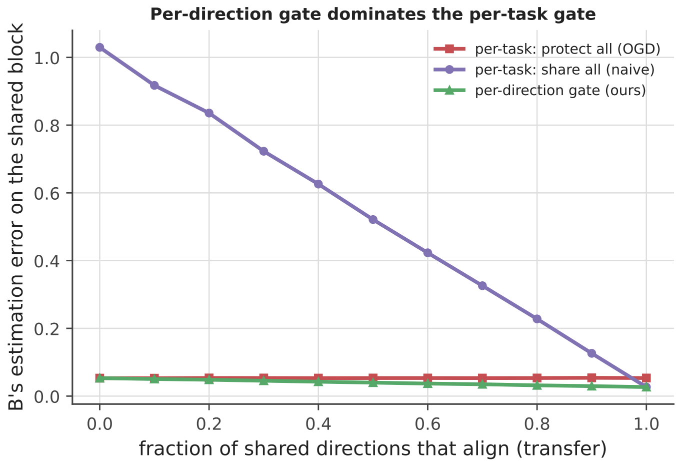

Figure S8: Per-direction vs per-task gating on a shared block, against the fraction of shared directions that align. The per-direction gate (green) follows the lower envelope—matching protect-all's robustness and share-all's transfer where each is right—while per-task share-all (purple) is catastrophic at low alignment.

igfa attains both the lowest average loss (4.28 vs naive 4.54, replay 4.59) and the least first-domain forgetting after the full stream (+0.80 vs naive +1.10, replay +0.91), carrying only a low-rank subspace as state. The domains are mutually dissimilar, so igfa here runs as pure structural projection. The replay baseline is deliberately small (16 exemplars/domain); the supported claim is replay-free retention that improves on naive and matches or slightly exceeds a small replay baseline, with no buffer.

Table S6: Continual pretraining of `pythia-1b` (1B parameters, 4-bit) + LoRA on six dissimilar text domains, boundary-free, with a Frequent-Directions-tracked subspace. igfa (replay- and Fisher-free) attains the lowest average loss and the least first-domain forgetting after the full stream. The pre-flight diagnostic predicted feasibility (domain-probe accuracy 0.85).

Method	Avg. final loss ↓	Domain-0 forgetting ↓	Extra state
Naive	4.54	+1.10	—
Replay (16/domain)	4.59	+0.91	buffer
igfa (FD, replay-free)	**4.28**	**+0.80**	subspace

S14 Extensions and generality

The interference functional applies beyond sequential continual learning. We give seven exact-A1 proofs-of-concept on synthetic constructions (the federated, out-of-distribution, task-free, and multimodal results average over 20 seeds; `evaluate_hypotheses.py`, `evaluate_applications.py`). These are directional demonstrations of scope, not benchmarks.

Federated learning is Σ-merging. Non-IID clients are tasks with different input geometries, and the weight divergence that FedAvg suffers coincides with the merge floor of Theorem 3. Replacing the FedAvg mean with the Σ-orthogonal aggregate $w_J = (\sum_i \Sigma_i)^+(\sum_i \Sigma_i w_i)$, where $(\cdot)^+$ is the Moore–Penrose pseudo-inverse (the Σ_i are low-rank, so $\sum_i \Sigma_i$ is generically singular and a plain inverse would not exist)—each client transmitting (Δ_i, Σ_i), or a low-rank sketch—lowers the mean client excess loss on non-IID data in *every* one of 20 seeds (mean 1.56×, minimum 1.14×). Federated aggregation and continual allocation are the same Σ-metric optimum.

The subspace summary is a free out-of-distribution signal. A minibatch whose gradient lies outside the occupied subspace U is structurally disjoint from the training distribution. The projection ratio $\|UU^\top g\|/\|g\|$ is 1.00 for in-distribution batches and 0.39 ± 0.05 for out-of-distribution batches (Cohen's $d \approx 18$, minimum 13 across seeds), so a single threshold flags OOD inputs with no auxiliary model—an uncertainty signal obtained directly from the continual-learning state.

Fully autonomous, task-free igfa. In a label-free stream the velocity of the recursive Gauss–Newton eigenbasis—the Frobenius commutator $\|U_t U_t^\top U_{t-1} U_{t-1}^\top - U_{t-1} U_{t-1}^\top U_t U_t^\top\|_F$—spikes exactly at distribution shifts, recovering randomly placed boundaries at recall and precision 1.00 over 20 streams (decay $\gamma = 0.6$). Composing this signal with the out-of-distribution guard above removes task labels entirely: the learner declares a boundary on a velocity spike, freezes the *pre-spike* subspace to protect the just-finished task, skips out-of-distribution/noise batches by their orthogonal-and-extreme gradient signature, and otherwise runs the gate. On clean streams this fully autonomous igfa *matches the oracle* provided with the true boundaries (excess loss 0.000 vs 0.000; naive 0.365); under injected extreme-magnitude OOD noise it holds at 0.033 while the same oracle—lacking the guard—is corrupted past 10^3, skipping 97% of noise batches at 0.05 false task-initializations per stream. Boundary detection, out-of-distribution rejection, and gated allocation are thus one self-regulating loop, with no human-provided task boundaries.

Multimodal interference is block-structured. For a joint embedding $\phi = [\phi_v, \phi_l]$, the second moment Σ_A partitions into intra-modal blocks Σ_v, Σ_l and a cross-modal block Σ_{vl}, and the interference energy $\frac{1}{2}\Delta^\top \Sigma_A \Delta$ distributes linearly across them. The cross-modal (alignment) interference of a fixed update scales with the off-diagonal $\|\Sigma_{vl}\|$—negligible when the modalities are unaligned and dominant when they are (energy ratio $\gg 1$ across all seeds) —so catastrophic decoupling is carried strictly by Σ_{vl}. A cross-modal igfa would Σ_{vl}-orthogonalize the alignment directions while sharing the intra-modal ones. This is a synthetic decomposition; a real multimodal encoder is needed to make it conclusive.

The functional governs physics-informed constraints. A differential constraint is a further quadratic form in feature space. For a physics-informed network with frozen features ϕ, the PDE residual acts on ϕ'', so the physics second moment is $\Sigma_{\mathrm{PDE}} = \mathbb{E}[\phi'' \phi''^\top]$ and a data-only update Δ raises the physics loss by exactly $\nabla L_{\mathrm{PDE}}^\top \Delta + \frac{1}{2}\Delta^\top \Sigma_{\mathrm{PDE}} \Delta$. On a 1D Poisson problem (sine random-feature basis) the interference energy $\frac{1}{2}\Delta^\top \Sigma_{\mathrm{PDE}} \Delta$ alone predicts the measured spike in the PDE residual at $r = 0.997$, and the full identity to machine precision ($r = 1.00000$, relative error 2×10^{-15}; Fig. S9a): the functional quantifies how empirical data corrupt a physical prior, with the only change being $\phi \mapsto \phi''$. Projecting the data gradient off the leading physics directions reduces the induced violation (3× here) but cannot eliminate it when Σ_{PDE} is full rank—an instance of the distortion floor of Theorem 4 under a shared active subspace.

Policy-gradient interference is governed by the Fisher. For a linear-Gaussian policy $\pi(a \mid s) = \mathcal{N}(\theta^\top s, \sigma^2)$ under reward $-(a - w^\top s)^2$ the expected return is $J(\theta) = -(\theta - w)^\top F(\theta - w) - \sigma^2$ with $F = \mathbb{E}[ss^\top]$ the empirical Fisher: the Σ of supervised interference is replaced verbatim by F. Across two control environments sharing a three-dimensional state subspace, routing the policy gradient orthogonal to the old-environment Fisher subspace retains the first environment perfectly (return 0.000 vs naive -10.9; a Fisher-weighted anchor, the EWC analogue, reaches -1.5; Fig. S9b) while solving the second in its free directions. The removability dichotomy of Corollary 1 holds unchanged in reinforcement learning; only the metric's estimator changes. This is the noise-free expected gradient—a stochastic PPO/MuJoCo test is left to future work.

The distortion floor can be trained away. Theorem 4's floor vanishes when task second moments are mutually orthogonal, a property a feature map can be optimised to enforce. Adding the penalty $\sum_{A<B} \mathrm{Tr}(\Sigma_A \Sigma_B)$ to a shared encoder drives its per-task subspaces apart: on overlapping synthetic tasks it cuts mean inter-task overlap 10.6× ($0.27 \to 0.03$) and the floor proxy $\mathrm{Tr}(\Sigma_A \Sigma_B)$ by over three orders of magnitude ($3.7 \to 10^{-3}$; Fig. S9c), leaving igfa to operate in the disjoint, lossless regime of Corollary 1. Structural allocation and representation learning are thus complementary: a backbone meta-trained for feature orthogonality removes the need for gating, so representation learning and structural allocation address the same distortion floor.

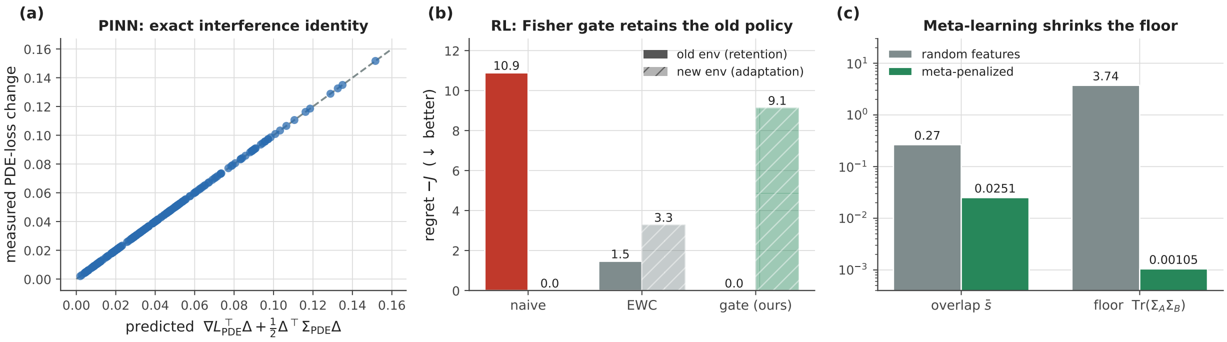

Figure S9: Three application domains for the interference functional (`evaluate_applications.py`). *(a)* Physics-informed regression: the interference identity with Σ_{PDE} predicts the measured PDE-loss change of data-only updates to machine precision (dashed: $y = x$). *(b)* Policy gradients: routing the expected policy gradient orthogonal to the old environment's Fisher subspace retains the old policy at zero regret while the new environment is learned in its free directions; naive descent forgets, and the Fisher-weighted anchor (the EWC analogue) is intermediate. *(c)* A feature map trained with the $\mathrm{Tr}(\Sigma_A \Sigma_B)$ penalty reduces inter-task overlap and the distortion-floor proxy by orders of magnitude (log scale), moving igfa into the lossless regime.

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